

NCEC ROM Map Editor

Detailed User Manual

Target ECU: Denso NC / Renesas SH7058 CALID LFG7EG Generation NC1

ROM: ncec_rom_dump.bin Generated: 2026-10-05

This document is generated automatically from the values in the actual ROM. Re-run tools/gen_manual.py after any change to the ROM or the tool to refresh it.

Safety notice

- Work on a spare (bench) ECU. Do not write to the ECU installed in a vehicle.
- Changing ignition timing, AFR or limiters can destroy the engine. Do not change anything you do not understand.
- Checksums are corrected automatically on save. Flashing the real ECU additionally requires an SBL (see below).

1. Overview

This is an open, self-built GUI tool that loads the ROM of a 2005 NCEC Roadster (LF-VE / SH7058 ECU), displays and edits its maps (tables), and saves with the checksum corrected. It aims to be a native-reflash tuning environment in the style of LinkECU / Haltech.

What it can do

- Load a ROM (.bin) and pick maps from a category tree (with localized names).
- Heatmap view and cell editing of 2D/3D maps, axis labels, and per-map notes.
- Automatic checksum correction on save (byte-for-byte verified against nc-flash).
- Live monitoring on the real ECU (OBD-II): RPM, load, coolant, voltage, DTCs, etc., with active-cell tracking of the current operating point.
- ECU read-out (Mode 23), ECU-vs-ROM compare, and ROM-vs-ROM compare.

Current limitations

- 'Flashing' the real ECU requires an SBL (Secondary Boot Loader), which is still being sourced. Authentication, the sequence, checksums and the rest are already implemented and verified; the tool can flash once an SBL is supplied.
- On a bench (engine off) the live operating point is static, so active-cell tracking stays pinned to the lower-left (it moves while driving or when injecting signals).

2. Using the Software

Screen layout

- Left: the category tree (localized). The search box filters by name / category.
- Top-right: info on the selected map (address, dimensions, unit, type) and a notes panel.
- Middle-right: the live value panel (while Live is running).
- Bottom-right: the heatmap grid (click a cell -> type a value -> Enter to commit).

Toolbar buttons

- Open ROM ... Load a ROM using the default (LibreTuner) definitions.
- Save / Save As ... Write out with the checksum corrected automatically.
- Help ... Quick help (docs/editor_help_ja.md).
- Live ▶/■ ... Connect to the real ECU over OBD and start/stop live view. The current operating point's cell is outlined in red.
- Read ECU ... Read the full 1 MB from the real ECU and save it as a .bin (read-only).
- Compare ECU ... Read the real ECU and diff it against the ROM you have open.
- Compare ROMs ... Diff two .bin files.

Basic workflow

(1) Open ROM -> (2) pick a map in the tree -> (3) edit cells (Enter to commit) -> (4) save (checksum auto-corrected). The heatmap colors show each value's relative position.

Cell colors and symbols

- Heatmap: low = blue -> high = red. It shows the relative distribution of values.
- '—' (gray): an unused cell (NaN, e.g. 0xFF).
- Red outline: the cell at the current operating point while Live is running.

Notes on live view and comparison

- Read / compare / Live each take exclusive use of the CAN adapter (gs_usb) — they cannot run at the same time.
- Read / compare perform authentication (programming session + security unlock) but are read-only (no erase/write), and return to the normal session afterwards.
- Doing this from a vehicle's OBD-II port requires GND (4/5) in addition to CAN-H (6) / CAN-L (14).

Flashing: current status

The full sequence — authentication -> RoutineControl -> RequestDownload (0x8000/0xFF800) -> SBL + main transfer (0x400 blocks) -> TransferExit -> reset — is implemented and verified on the real ECU up to the step before writing. Only the SBL (for NC1, 0x1800 bytes) is still missing, so actual writing is on hold.

3. Map Details and How to Tune Them

This chapter covers the most representative and important maps, with their current values from the actual ROM. (Every table is browsable from the editor's tree.)

Spark Base (Base Ignition Advance)

Table: Spark Base - High Fuel Request | High-Det, IMRC @0xCE590 (category: Spark Base - High Fuel Request)

- Overview: Base ignition timing (advance) versus RPM x load. The single most important map for torque, fuel economy and knock margin. The usual shape pulls advance back at high load and builds it up in the low-to-mid load range.

Spark Base (Base Ignition Advance)

Engine RPM (rpm) →								
↓ Engine Load	500	1000	1500	2000	2500	3000	3500	4000
0	16	25	37	45	58	60	46	45
0	16	25	37	44	51	53	46	45
0	16	24	36	44	46	49	44	42
0	15	23	35	40	42	45	42	40
0	8	17	25	33	39	41	37	37
0	2	11	19	25	27	39	35	35
0	-2	7	13	21	23	35	34	33
1	-5	4	10	13	21	28	31	30
1	-9	2	5	10	18	23	24	26
1	-15	-4	2	7	12	17	21	23
1	-16	-9	-0	4	10	16	19	20
1	-17	-10	-1	2	8	13	16	17
1	-18	-11	-2	2	4	12	14	16
1	-19	-12	-2	0	4	11	13	14
1	-20	-13	-3	-0	3	10	12	13

- Effect of changes: Higher (more advance) = more torque / economy but higher knock risk. Lower (retard) = safer but less power. The high-load / low-RPM corner knocks most easily, so raise it only 1-2 deg at a time while monitoring for knock.

⚠ Caution: Running premium-fuel values on regular fuel causes knock and engine damage. Any change beyond stock +2 deg must be paired with a road knock log (or live knock-sensor monitoring).

Ignition Coil Dwell Time

Table: Ignition Coil - Dwell Time | Speed Limit Inactive @0xBBAD0 (category: Ignition Coil Dwell Time)

- Overview: Coil charge (dwell) time in ms. Versus RPM x voltage it sets the time needed to build the required spark energy.

Ignition Coil Dwell Time

↓ Module V (V)	Engine RPM (rpm) →												
	500.00	1000.00	1500.00	2000.00	2500.00	3000.00	3500.00	4000.00	4500.00	5000.00	5500.00	6000.00	6500.00
6.50	9.50	9.50	7.02	7.02	7.02	4.63	4.63	4.63	4.63	4.63	4.02	4.02	4.02
7.75	7.90	7.90	5.95	5.95	5.95	4.06	4.06	4.06	4.06	4.06	3.50	3.50	3.50
9.00	7.07	7.07	4.88	4.88	4.88	3.96	3.96	3.96	3.78	3.78	3.39	3.39	3.24
10.25	4.91	4.91	4.91	3.73	3.73	3.22	3.22	3.22	3.22	3.22	3.08	3.08	3.08
11.50	4.22	4.22	4.22	4.22	3.21	2.77	2.77	2.77	2.77	2.77	2.77	2.77	2.77
12.75	3.64	3.64	3.64	3.64	3.64	2.66	2.66	2.66	2.66	2.66	2.66	2.66	2.66
14.00	3.13	3.13	3.13	3.13	3.13	3.13	2.48	2.48	2.48	2.48	2.48	2.48	2.48
15.25	2.77	2.77	2.77	2.77	2.77	2.77	2.33	2.33	2.33	2.33	2.33	2.33	2.33
16.50	2.47	2.47	2.47	2.47	2.47	2.47	2.47	2.15	2.15	2.15	2.15	2.15	2.15

- Effect of changes: Tune this when changing coils (aftermarket, GDI coils, etc.). Too short causes misfire at high RPM / high load; too long overheats the coil and shortens its life. With stock coils it generally needs no change.

⚠ Caution: Do not blindly extend dwell across the board (overheating destroys the coil). Base any change on the coil maker's recommended values.

Target AFR, Open Loop (Fuel Target OL)

Table: Fuel Target OL (Lambda) | APP Above WOT Threshold, High Det @0xCD13C (category: Fuel Target OL - Base)

- Overview: Target air-fuel ratio (lambda / AFR) at high load (open loop). In regions such as full-throttle acceleration the ECU leaves O2 feedback and fuels to this target.

Target AFR, Open Loop (Fuel Target OL)

↓ Throttle (TP) (%)	Engine RPM (rpm) →												
	500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500
12	13	13	22	22	22	13	13	13	19	22	24	31	36
25	13	13	22	22	22	13	13	13	19	22	24	31	36
38	13	13	22	22	22	13	13	13	19	22	24	31	36
50	13	13	22	22	22	13	13	13	19	22	24	31	36
62	13	13	22	22	22	13	13	13	19	22	24	31	36
75	13	13	22	22	22	13	13	13	19	22	24	31	36
88	13	13	22	22	22	13	13	13	19	22	24	31	36
100	13	13	22	22	22	13	13	13	19	22	24	31	36

- Effect of changes: Richer = lower combustion temperature, safer and power-oriented (power AFR is roughly lambda 0.85-0.88). Leaner = economy but dangerous at high load. Keeping the high-load region on the rich side is the safe choice.

⚠ Caution: A lean mixture at high load risks immediate knock / melted pistons. Keep any leaning minimal and verify it with a measured wideband AFR.

Target AFR, Closed Loop (Fuel Target CL)

Table: Fuel Target CL - Base (Lambda) @0xD4288 (category: Fuel Target CL - Base (no patch required))

- Overview: Target air-fuel ratio in closed loop (normally stoichiometric, near lambda 1). The O2 sensor keeps the mixture on this target.

Target AFR, Closed Loop (Fuel Target CL)

	Engine RPM (rpm) →									
↓ Engine Load	500.00	1000.00	1500.00	2000.00	2500.00	3000.00	3500.00	4000.00	4500.00	5000.00
0.12	0.50	0.58	0.47	0.35	0.57	0.30	0.31	0.33	0.35	0.36
0.19	0.50	0.50	0.38	0.26	0.46	0.31	0.32	0.34	0.36	0.38
0.25	0.50	0.42	0.30	0.17	0.36	0.32	0.33	0.35	0.37	0.39
0.31	0.35	0.35	0.24	0.12	0.25	0.33	0.34	0.36	0.38	0.40
0.38	0.27	0.27	0.17	0.08	0.19	0.40	0.34	0.34	0.39	0.39
0.44	0.28	0.28	0.17	0.06	0.21	0.43	0.32	0.33	0.35	0.37
0.50	0.29	0.28	0.17	0.03	0.23	0.45	0.36	0.37	0.39	0.41
0.56	0.26	0.25	0.17	0.20	0.22	0.57	0.57	0.55	0.63	0.66
0.62	0.23	0.23	0.02	0.30	0.35	0.55	0.71	0.73	0.79	0.83
0.69	0.20	0.20	0.21	0.35	0.35	0.65	0.86	0.88	0.95	1.00
0.75	0.17	0.17	0.16	0.40	0.45	0.70	0.91	0.95	1.00	1.05
0.81	0.14	0.14	0.23	0.49	0.60	0.85	0.98	1.03	1.09	1.14
0.88	0.11	0.11	0.29	0.59	0.86	0.96	1.14	1.19	1.25	1.32

- Effect of changes: Normally needs no change. It is central to catalyst protection and emissions, so leave it alone unless you have a specific reason to move off stoichiometric.

⚠ Caution: Moving away from lambda 1 worsens emissions, degrades the catalyst and can trigger the check-engine light. Leave it as-is as a rule.

Variable Valve Timing (VVT)

Table: VCT Target @0xBADC4 (category: Variable Cam Timing - Base)

- Overview: Target intake (or exhaust) cam advance versus RPM x load. It shapes where torque appears (low-end vs top-end) and affects volumetric efficiency and internal EGR.

Variable Valve Timing (VVT)

↓ Engine Load	Engine RPM (rpm) →												
	500	1000	1500	2000	2500	3000	3500	4000	4500	5000	5500	6000	6500
0	0	0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	0	0	20	20	0	0	0	0	0	0	0
0	0	0	20	20	20	20	0	0	0	0	0	0	0
0	0	0	20	20	20	20	0	0	0	0	0	0	0
0	0	0	20	20	20	20	15	0	0	0	0	0	0
0	0	0	20	20	20	20	15	15	0	0	0	0	0
0	0	0	20	20	20	20	15	15	0	0	0	0	0
1	0	0	20	20	20	20	15	15	0	0	0	0	0
1	0	0	25	20	20	25	15	15	0	0	0	0	0
1	0	0	25	25	25	25	25	20	0	0	0	0	0
1	0	0	25	25	25	25	25	25	15	10	5	10	5
1	0	0	25	25	25	25	25	25	25	20	15	10	5
1	0	0	25	25	25	25	25	25	25	20	15	10	5
1	0	0	25	25	25	25	25	25	25	20	15	10	5
1	0	0	25	25	25	25	25	25	25	20	15	10	5

- Effect of changes: You can shape the character, e.g. adding advance to target mid-range torque. The effect is easy to feel, but the optimum must be found by measurement.

⚠ Caution: Excessive advance causes valve-train problems, knock and unstable idle. Move in small steps. Do not touch the 'DO NOT MODIFY' tables.

Rev Limiter (Rev Limit)

- Overview: Upper-RPM protection. Depending on conditions it uses either fuel cut or throttle cut.
 - Fuel cut: warm / MT (the effective rev limit): 7,500 (@0xC487C)
 - Fuel cut: AT (while the 73C0 timer is active): 6,500 (@0xC4878)
 - Fuel cut: cold / below primary threshold / fault: 5,500 (@0xC4870)
 - Fuel cut: cold / secondary / cranking: 5,500 (@0xC4874)
 - Throttle cut: clutch depressed (neutral free-rev protection): 6,500 (@0xC53BC)
 - Throttle cut: clutch released (driving / effectively disabled): 10,000 (@0xC53C0)
- Effect of changes: Raising the value raises the allowed RPM. If you raise it, always confirm the valve-train and drivetrain are rated for that speed.

⚠ Caution: Settings beyond the mechanical limit lead straight to engine failure. Do not raise it carelessly.

Speed Limiter / Idle (Speed & Idle)

- Overview: The conditions that trigger the vehicle-speed limiter, plus representative target-idle values.
 - Speed limiter: activation vehicle-speed threshold: 300 (@0xC47EC)
 - Speed limiter: activation RPM threshold: 4,000 (@0xC47E4)
- Effect of changes: Removing the speed limiter is at your own risk; obey the law on public roads. Higher idle = more stable, lower idle = quieter (too low stalls).

⚠ Caution: Check the intended use and the law before changing the speed limiter. Also confirm cold-start stability when changing idle.

4. Full Map Reference (all tables, by category)

This chapter lists every table in the definition (875 total). Each category carries a description and an 'Effect of changes' note, and every table shows its [dimensions], axes, value range (or the value itself for 1x1) and address. For in-depth notes and tuning guidance see Chapter 3 (key maps).

DBW - Cruise Control

Description: Electronic throttle (drive-by-wire) mapping/response.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Cruise Control Correction - Base [6×9] X=Vehicle Speed (km/h) Y=Gear range 30–90 [0xBA744]
Cruise Control Correction Sensitivity - 1st Gear [1×39] X=Vehicle Speed (km/h) range -6–0.15 [0xB9BB0]
Cruise Control Correction Sensitivity - 2nd Gear [1×39] X=Vehicle Speed (km/h) range -6–0.15 [0xB9CE8]
Cruise Control Correction Sensitivity - 3rd Gear [1×39] X=Vehicle Speed (km/h) range -6–0.15 [0xB9E20]
Cruise Control Correction Sensitivity - 4th Gear [1×39] X=Vehicle Speed (km/h) range -6–0.15 [0xB9F58]
Cruise Control Correction Sensitivity - 5th Gear [1×39] X=Vehicle Speed (km/h) range -6–0.15 [0xBA090]
Cruise Control Correction Sensitivity - 6th Gear [1×39] X=Vehicle Speed (km/h) range -6–0.15 [0xBA1C8]
Cruise Control Correction Sensitivity - Multiplier to Percentage [6×9] X=Vehicle Speed (km/h) Y=Gear range 100–100 [0xBA858]
Cruise Control Correction Sensitivity - Non Gear [1×39] X=Vehicle Speed (km/h) range 0–0 [0xB9A78]
Cruise Control Sensitivity - 1st Gear [1×39] X=Vehicle Speed (km/h) range -95.73–95.73 [0xB9328]
Cruise Control Sensitivity - 2nd Gear [1×39] X=Vehicle Speed (km/h) range -95.73–95.73 [0xB9460]
Cruise Control Sensitivity - 3rd Gear [1×39] X=Vehicle Speed (km/h) range -95.73–95.73 [0xB9598]
Cruise Control Sensitivity - 4th Gear [1×39] X=Vehicle Speed (km/h) range -95.73–95.73 [0xB96D0]
Cruise Control Sensitivity - 5th Gear [1×39] X=Vehicle Speed (km/h) range -95.73–95.73 [0xB9808]
Cruise Control Sensitivity - 6th Gear [1×39] X=Vehicle Speed (km/h) range -95.73–95.73 [0xB9940]
Cruise Control Sensitivity - Non Gear [1×39] X=Vehicle Speed (km/h) range 0–0 [0xB91F0]
Cruise Control Target APP - Base [6×9] X=Vehicle Speed (km/h) Y=Gear range 30–90 [0xBA2A0]

DBW - Throttle Angle Commanded Conversion

Description: Electronic throttle (drive-by-wire) mapping/response.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Throttle Angle - Maximum = 85 [0xBB2CC]
Throttle Percentage to Throttle Angle Multiplier = 0.85 [0xBB0D4]

DBW - Throttle Duty Desired

Description: Electronic throttle (drive-by-wire) mapping/response.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

APP to Throttle Duty Desired - WIP, Brake On, VSS LT 2kph, AT [1×18] X=Pedal (APP) (%) range 0–100 [0xC7E9C]

DBW - Throttle Duty Desired Comp - Aggregate

Description: Electronic throttle (drive-by-wire) mapping/response.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Estimated Volumetric Flow - Default | Fault = 0.2 [0xD6F44]
Estimated Volumetric Flow - High Elevation Activation Threshold = 19266 [0xD7CF8]
Estimated Volumetric Flow - Maximum [1×16] X=Coolant (°C) range 30–60 [0xBB788]
Estimated Volumetric Flow - Minimum [1×16] X=Coolant (°C) range 0–4.5 [0xBB808]
Non Emissions Estimated Mass Air-Flow - Maximum | RPM GT Idle Speed = 18500 [0xD6F80]

DTC Flags B

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

C0073 - CAN system communication error = 0 [0xDD2C9]
 C0101 - Communication error to TCM = 1 [0xDD2CA]
 C0121 - Communication error to ABS HU/CM or DSC HU/CM = 1 [0xDD2CB]
 C0155 - Communication error to instrument cluster A = 1 [0xDD2CC]
 C0155 - Communication error to instrument cluster B = 0 [0xDD2CD]
 P0011 - CMP timing over-advanced = 1 [0xDD2A7]
 P0012 - CMP timing over-retarded = 1 [0xDD2A8]
 P0016 - CKP-CMP correlation = 1 [0xDD298]
 P0030 - A/F sensor heater control circuit problem = 1 [0xDD268]
 P0031 - A/F sensor heater circuit low input = 1 [0xDD269]
 P0032 - A/F sensor heater circuit high input = 1 [0xDD26A]
 P0037 - HO2S heater circuit low input = 1 [0xDD26B]
 P0038 - HO2S heater circuit high input = 1 [0xDD26C]
 P0069 - MAP (manifold) vs BARO (atmospheric) correlation problem = 1 [0xDD27C]
 P0101 - MAF sensor circuit range / performance problem = 1 [0xDD276]
 P0102 - MAF sensor circuit low input = 1 [0xDD277]
 P0103 - MAF sensor circuit high input = 1 [0xDD278]
 P0107 - MAP sensor circuit low input = 1 [0xDD27D]
 P0108 - MAP sensor circuit high input = 1 [0xDD27E]
 P0111 - IAT sensor circuit range / performance problem = 1 [0xDD279]
 P0112 - IAT sensor circuit low input = 1 [0xDD27A]
 P0113 - IAT sensor circuit high input = 1 [0xDD27B]
 P0116 - ECT circuit range / performance problem = 1 [0xDD26E]
 P0117 - ECT sensor circuit low input = 1 [0xDD26F]
 P0118 - ECT sensor circuit high input = 1 [0xDD270]
 P0122 - TP sensor no.1 circuit low input = 1 [0xDD286]
 P0123 - TP sensor no.1 circuit high input = 1 [0xDD287]
 P0125 - Excessive time to enter closed loop fuel control = 1 [0xDD271]
 P0126 - Coolant thermostat stuck open A = 1 [0xDD272]
 P0128 - Coolant thermostat stuck open B = 1 [0xDD273]
 P0130 - A/F sensor circuit problem = 1 [0xDD25D]
 P0131 - A/F sensor circuit low input = 1 [0xDD25E]
 P0132 - A/F sensor circuit high input = 1 [0xDD25F]
 P0133 - A/F sensor circuit problem = 1 [0xDD260]
 P0134 - A/F sensor no activity detected = 1 [0xDD261]
 P0137 - HO2S circuit low input = 1 [0xDD264]
 P0138 - HO2S circuit high input = 1 [0xDD265]
 P0139 - N/A HO2S circuit slow response = 1 [0xDD266]
 P0140 - HO2S no activity detected = 1 [0xDD267]
 P0222 - TP sensor no.2 circuit low input = 1 [0xDD288]
 P0223 - TP sensor no.2 circuit high input = 1 [0xDD289]
 P0300 - Random misfire detected A = 1 [0xDD250]
 P0300 - Random misfire detected B = 1 [0xDD251]
 P0327 - KS circuit low input = 1 [0xDD281]
 P0328 - KS circuit high input = 1 [0xDD282]
 P0335 - CKP sensor circuit problem = 1 [0xDD296]
 P0340 - CMP sensor circuit problem = 1 [0xDD297]
 P0351 - N/A Ignition Coil Primary Circuit Fault - Cylinder 1 = 1 [0xDD299]
 P0352 - N/A Ignition Coil Primary Circuit Fault - Cylinder 2 = 1 [0xDD29A]
 P0353 - N/A Ignition Coil Primary Circuit Fault - Cylinder 3 = 1 [0xDD29B]
 P0354 - N/A Ignition Coil Primary Circuit Fault - Cylinder 4 = 1 [0xDD29C]
 P0401 - EGR flow insufficient detected = 1 [0xDD26D]
 P0403 - EGR valve (stepping motor) circuit problem = 1 [0xDD2A4]
 P0421 - Warm-up catalyst system efficiency below threshold = 1 [0xDD252]
 P0441 - EVAP system incorrect purge flow = 1 [0xDD253]
 P0442 - EVAP system leak detected (small leak) = 1 [0xDD254]
 P0443 - Purge solenoid valve circuit = 1 [0xDD29D]
 P0446 - Change over valve (EVAP system leak detection pump) stuck close = 1 [0xDD29E]
 P0455 - EVAP system leak detected (gross leak / no flow) = 1 [0xDD255]
 P0456 - EVAP system leak detected (very small leak) = 1 [0xDD256]

P0461 - Fuel gauge range / performance problem = 1 [0xDD283]
 P0462 - Fuel gauge sender unit circuit low input = 1 [0xDD284]
 P0463 - Fuel gauge sender unit circuit high input = 1 [0xDD285]
 P0480 - Cooling fan relay no.1 control circuit malfunction = 0 [0xDD2BE]
 P0481 - Cooling fan relay no.2 control circuit malfunction = 0 [0xDD2BF]
 P0482 - Cooling fan relay no.3 control circuit malfunction = 0 [0xDD2C0]
 P0500 - VSS circuit problem = 1 [0xDD295]
 P0505 - Idle speed control system problem = 0 [0xDD2C7]
 P0506 - Idle speed control system RPM lower than expected = 1 [0xDD2B4]
 P0507 - Idle speed control system RPM higher than expected = 1 [0xDD2B5]
 P050A - Cold start idle air control system performance problem = 1 [0xDD274]
 P050B - Cold start ignition timing performance problem = 1 [0xDD275]
 P0550 - PSP switch circuit malfunction = 1 [0xDD291]
 P0564 - Cruise control switch circuit malfunction = 0 [0xDD2C4]
 P0571 - Brake switch circuit problem = 0 [0xDD2C5]
 P0601 - PCM memory check sum error = 1 [0xDD2B9]
 P0602 - PCM programming error = 1 [0xDD2B7]
 P0604 - PCM random access memory (RAM) error = 1 [0xDD2BA]
 P0606 - PCM processor error = 1 [0xDD2BB]
 P0610 - PCM vehicle options error = 1 [0xDD2B8]
 P0638 - Throttle valve actuator control circuit range/performance problem = 1 [0xDD2AD]
 P0661 - Variable intake air solenoid valve circuit low input = 0 [0xDD2BC]
 P0662 - Variable intake air solenoid valve circuit high input = 0 [0xDD2BD]
 P0703 - Brake switch input circuit problem = 1 [0xDD292]
 P0704 - CPP switch input circuit problem = 1 [0xDD293]
 P0850 - Neutral switch input circuit problem = 1 [0xDD294]
 P1260 - Immobilizer switch problem = 0 [0xDD2C6]
 P2004 - Variable tumble shutter valve stuck open = 1 [0xDD2A9]
 P2006 - Variable tumble valve stuck closed = 1 [0xDD2AA]
 P2009 - Variable tumble solenoid valve circuit low input = 1 [0xDD2AB]
 P2010 - Variable tumble solenoid valve circuit high input = 1 [0xDD2AC]
 P2088 - OCV circuit low input = 1 [0xDD2A5]
 P2089 - OCV circuit high input = 1 [0xDD2A6]
 P2096 - Target A/F feedback system too lean = 1 [0xDD25B]
 P2097 - Target A/F feedback system too rich = 1 [0xDD25C]
 P2101 - throttle valve actuator circuit range/performance problem = 1 [0xDD2AE]
 P2107 - Throttle valve actuator control module processor error = 1 [0xDD2AF]
 P2108 - Throttle valve actuator control module performance error = 1 [0xDD2B0]
 P2109 - TP sensor minimum stop range/performance problem = 1 [0xDD2B1]
 P2112 - Throttle valve actuator control system range/performance problem = 1 [0xDD2B2]
 P2119 - Throttle valve actuator control throttle body range/performance problem = 1 [0xDD2B3]
 P2122 - APP sensor no.1 circuit low input = 1 [0xDD28B]
 P2123 - APP sensor no.1 circuit high input = 1 [0xDD28C]
 P2127 - APP sensor no.2 circuit low input = 1 [0xDD28D]
 P2128 - APP sensor no.2 circuit high input = 1 [0xDD28E]
 P2135 - TP sensor no.1/no.2 voltage correlation problem = 1 [0xDD28A]
 P2138 - APP sensor no.1/no.2 voltage correlation problem = 1 [0xDD28F]
 P2177 - System Too Lean Off Idle = 1 [0xDD257]
 P2178 - System Too Rich Off Idle = 1 [0xDD258]
 P2187 - System Too Lean At Idle = 1 [0xDD259]
 P2188 - System Too Rich At Idle = 1 [0xDD25A]
 P2195 - N/A HO2S stuck lean = 1 [0xDD262]
 P2196 - N/A HO2S stuck rich = 1 [0xDD263]
 P2228 - BARO sensor circuit low input = 1 [0xDD27F]
 P2229 - BARO sensor circuit high input = 1 [0xDD280]
 P2401 - EVAP system leak detection pump motor circuit low input = 1 [0xDD29F]
 P2402 - EVAP system leak detection pump motor circuit high input = 1 [0xDD2A0]
 P2404 - EVAP system leak detection pump sense circuit problem = 1 [0xDD2A1]
 P2405 - EVAP system leak detection pump sense circuit low input = 1 [0xDD2A2]
 P2407 - EVAP system leak detection pump sense circuit intermittent = 1 [0xDD2A3]

P2502 - Charging system voltage problem = 0 [0xDD2C1]
P2503 - Charging system voltage low input = 0 [0xDD2C2]
P2504 - Charging system voltage high input = 0 [0xDD2C3]
P2507 - PCM battery voltage low input = 1 [0xDD290]
P2610 - PCM internal engine off timer performance problem = 1 [0xDD2B6]
P9342 - N/A = 0 [0xDD2C8]

DTC Thresholds

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

AFS cruise max load bound for charging efficiency A57A0 [1×13] X=Engine RPM (rpm) range 0.569–0.628 [0xDAC48]
AFS cruise min load bound for charging efficiency A5794 [1×13] X=Engine RPM (rpm) range 0.2–0.35 [0xDABE0]
DTC P0011 MONITOR CMP Timing Over Advanced Timer [1×8] X=Engine RPM (rpm) range 84–1250 [0xDC768]
DTC P0016 - CMP Out of Sync Counter Limit [1×8] X=Engine RPM (rpm) range 84–1250 [0xDC41C]
DTC P0016 - CPK-CMP Sync Offset [1×13] X=Engine RPM (rpm) range 8.757–11.192 [0xDC460]
DTC P0125 Excessive Time To Enter CL ECT Threshold [1×7] X=Coolant (°C) range 10–10 [0xDC28C]
DTC P0126 Monitor Coolant Thermostat Stuck Open Activation Timer [1×11] X=Coolant (°C) range 0–16900 [0xDB5AC]
DTC P0134 air fuel no activity detected - AFR Element Impedence Threshold [1×9] X=Coolant (°C) range 30000–80000 [0xD C8A0]
DTC P0134 air fuel no activity detected - Sub for Low ECT Start [1×9] X=Module V (V) range 21.56–100 [0xDC8E8]
DTC P0171 Fuel Trim Too Lean - Summed LT/STFT Lean Threshold [1×7] X=Intake Air (°C) range 30–30 [0xDA644]
DTC P0172 Fuel Trim Too Rich - Summed LT/STFT Rich Threshold [1×7] X=Intake Air (°C) range -30–30 [0xDA60C]
DTC P0421 Monitor Warm Up CAT Load Threshold - Max [1×13] X=Engine RPM (rpm) range 0.2968–0.48438 [0xDA79C]
DTC P0421 Monitor Warm Up CAT Load Threshold - Min [1×13] X=Engine RPM (rpm) range 0.1484–0.2031 [0xDA734]
DTC P0601 APP to TP Fault 20CC Activation Count [1×4] X=Throttle (TP) (%) range 10–40 [0xF4984]
DTC P0601 Minimum Requested Torque Threshold WIP [1×12] X=Engine RPM (rpm) range 2.5–2.5 [0xF49BC]
DTC P0601 Req Torque Fault 20C6 Activation Count A [1×8] X=Request range 40–40 [0xF488C]
DTC P0601 Req Torque Fault 20C6 Activation Count B [1×8] X=Request range 40–40 [0xF48BC]
DTC P0601 Req Torque Fault 20CA Activation Count A [1×12] X=Engine RPM (rpm) range 40–40 [0xF4914]
DTC P0601 Req Torque Fault 20CA Activation Count B [1×12] X=Engine RPM (rpm) range 40–40 [0xF495C]
DTC P0601 Requested Torque Delta Threshold [1×12] X=Engine RPM (rpm) range 147.07–258.65 [0xF4A1C]
DTC P0601 Requested Torque Detla Threshold DUPE [1×12] X=Engine RPM (rpm) range 147.07–258.65 [0xF4A7C]
DTC P0601 TP Comp Fault 20C8 Activation Count [1×4] X=Throttle (TP) (%) range 10–32 [0xF48DC]
DTC P0638 FAULT 20BC TP Threshold [1×4] X=Throttle (TP) (%) range 5.6–22 [0xDCA00]
DTC P0638 TP GT Closed Min Activation Timer [1×4] X=Throttle (TP) (%) range 31–100 [0xDC9D0]
DTC P0638 TP LT Closed MIN Activation Timer [1×4] X=Throttle (TP) (%) range 125–125 [0xDC9E8]
DTC P2135 TP Sensor 2 Offset (ZEROED) [1×7] X=Throttle (TP) (%) range 0–0 [0xDCB68]
TP factor for Estimated MAF [1×15] X=Engine RPM (rpm) range 4.39–59.74 [0xDC334]

Data Integrity Checks - DBW Throttle Duty Commanded Limits

Description: ETC/torque safety-monitor plausibility (fail-safe). Keep in sync with the control maps.

Effect of changes: Raise together with the control map or the ETC monitor trips fail-safe.

DISABLED - Throttle Duty Commanded Scaling (Bypass Torque Request) [1×9] X=Pedal (APP) (%) range 0–100 [0xF7BDC]
Throttle Duty Commanded - Maximum [1×9] X=Pedal (APP) (%) range 16.5–100 [0xF7C24]
Throttle Duty Commanded - Minimum [1×9] X=Pedal (APP) (%) range 0–0 [0xF7C6C]

Data Integrity Checks - DBW Throttle Duty Desired

Description: ETC/torque safety-monitor plausibility (fail-safe). Keep in sync with the control maps.

Effect of changes: Raise together with the control map or the ETC monitor trips fail-safe.

APP to Throttle Duty Desired - 1st Gear [10×18] X=Pedal (APP) (%) Y=Engine RPM (rpm) range 0–100 [0xF7C90]
APP to Throttle Duty Desired - 2nd Gear [10×18] X=Pedal (APP) (%) Y=Engine RPM (rpm) range 0–100 [0xF7F60]
APP to Throttle Duty Desired - 3rd Gear [10×18] X=Pedal (APP) (%) Y=Engine RPM (rpm) range 0–100 [0xF8230]
APP to Throttle Duty Desired - 4th, 5th, 6th Gear [10×18] X=Pedal (APP) (%) Y=Engine RPM (rpm) range 0–100 [0xF8500]
APP to Throttle Duty Desired - Max [1×18] X=Pedal (APP) (%) range 16.5–100 [0xF7930]
APP to Throttle Duty Desired - Min [1×18] X=Pedal (APP) (%) range 0–0 [0xF79C0]
APP to Throttle Duty Desired - Neutral [10×18] X=Pedal (APP) (%) Y=Engine RPM (rpm) range 0–100 [0xF87D0]
APP to Throttle Duty Desired - WIP | Brake On, VSS LT 2kph, AT [1×18] X=Pedal (APP) (%) range 0–100 [0xF78A0]

Data Integrity Checks - DBW Throttle Duty Desired + Comp to Torque Request

Description: ETC/torque safety-monitor plausibility (fail-safe). Keep in sync with the control maps.

Effect of changes: Raise together with the control map or the ETC monitor trips fail-safe.

Torque Request - Base (Brake Mean Effective Pressure in kpa) [30×16] X=Engine RPM (rpm) Y=Pedal (APP) (%) range -277.927 -1214.27 [0xF9254]

Torque Request - Limiter (Brake Mean Effective Pressure in kpa) | Fault 71B0 [30×16] X=Engine RPM (rpm) Y=Pedal (APP) (%) range -278.085-1168.97 [0xFA838]

Torque Request - Limiter (Brake Mean Effective Pressure in kpa) | Normal [30×16] X=Engine RPM (rpm) Y=Pedal (APP) (%) range -277.927-1214.27 [0xFA000]

Torque Request - Vehicle Speed Multiplier | AT [27×15] X=Pedal (APP) (%) Y=Vehicle Speed (km/h) range 1-1 [0xF8B48]

Data Integrity Checks - DBW Throttle Duty Desired Comp - Aggregate

Description: ETC/torque safety-monitor plausibility (fail-safe). Keep in sync with the control maps.

Effect of changes: Raise together with the control map or the ETC monitor trips fail-safe.

Estimated Volumetric Flow - Air Density Scaler [1×8] X=Baro (kPa) range 1-1 [0xF50D4]

Estimated Volumetric Flow - Default | KOEO [1×7] X=Coolant (°C) range 18500-18500 [0xF5098]

Estimated Volumetric Flow to Throttle Duty Aggregate Compensation [1×24] X=Est. Volumetric Flow (raw) range 0-13.0691 [0xF4B0C]

Data Integrity Checks - Idle Speed - Base

Description: ETC/torque safety-monitor plausibility (fail-safe). Keep in sync with the control maps.

Effect of changes: Raise together with the control map or the ETC monitor trips fail-safe.

Idle Speed - Base | In-Gear [1×8] X=Const range 720-1150 [0xF4DEC]

Idle Speed - Base | Not In-Gear [1×8] X=Const range 720-1750 [0xF4E2C]

Data Integrity Checks - Idle Speed Comp

Description: ETC/torque safety-monitor plausibility (fail-safe). Keep in sync with the control maps.

Effect of changes: Raise together with the control map or the ETC monitor trips fail-safe.

Idle Speed Comp - Start-up Base (ZEROED) | In-Gear, 96F4 Set [1×7] X=Coolant (°C) range 0-0 [0xF5028]

Idle Speed Comp - Start-up Base Limit (ZEROED) | In-Gear [1×7] X=Coolant (°C) range 0-0 [0xF4F80]

Idle Speed Comp - Start-up Base Limit (ZEROED) | Not In-Gear [1×7] X=Coolant (°C) range 0-0 [0xF4FF0]

Idle Speed Comp - Start-up Base | In-Gear [1×7] X=Coolant (°C) range 0-20 [0xF4F48]

Idle Speed Comp - Start-up Base | Not In-Gear [1×7] X=Coolant (°C) range 0-675 [0xF4FB8]

Idle Speed Comp - Start-up Base | Not In-Gear, 96F4 Set [1×7] X=Coolant (°C) range 0-350 [0xF5060]

Idle Speed Comp - Start-up Multiplier | In-Gear [1×17] X=Throttle (%) range 0-1 [0xF4D00]

Idle Speed Comp - Start-up Multiplier | Not In-Gear [1×17] X=Throttle (%) range 0-1 [0xF4D88]

Data Integrity Checks - Spark Idle Load - AC Comp

Description: ETC/torque safety-monitor plausibility (fail-safe). Keep in sync with the control maps.

Effect of changes: Raise together with the control map or the ETC monitor trips fail-safe.

AC Comp Base | AC, RO2S Heater Off, In-Gear [1×8] X=Coolant (°C) range 0.02-0.085 [0xF5404]

AC Comp Base | AC, RO2S Heater Off, Not In-Gear [1×8] X=Coolant (°C) range 0.02-0.07 [0xF5444]

AC Comp Base | AC, RO2S Heater On, In-Gear [1×8] X=Coolant (°C) range 0.02-0.06 [0xF5484]

AC Comp Base | AC, RO2S Heater On, Not In-Gear, [1×8] X=Coolant (°C) range 0.02-0.05 [0xF54C4]

Data Integrity Checks - Spark Idle Load - Alternator Comp

Description: ETC/torque safety-monitor plausibility (fail-safe). Keep in sync with the control maps.

Effect of changes: Raise together with the control map or the ETC monitor trips fail-safe.

Alt Comp - Alt to Ctrl Mod Delta Hysteresis [1×11] X=(unused) range 7.8125-7.8125 [0xF5708]

Alt Comp - Alternator to Estimated MAF Add | In-Gear [1×11] X=Alternator range 0-2.2 [0xF5550]

Alt Comp - Alternator to Estimated MAF Add | Not In-Gear [1×11] X=Alternator range 0-2.2 [0xF55A8]

Alt Comp Base | Alternator Field Control Duty GT 75 [1×11] X=Interp range 0-0.22 [0xF5600]

Alt Comp Base | In-Gear, Neg Alt to Ctrl Mod Delta [1×11] X=Interp range 0-0.1 [0xF5658]

Alt Comp Base | Not In-Gear, Neg Alt to Ctrl Mod Delta [1×11] X=Interp range 0-0.016 [0xF56B0]

Data Integrity Checks - Spark Idle Load - Base

Description: ETC/torque safety-monitor plausibility (fail-safe). Keep in sync with the control maps.

Effect of changes: Raise together with the control map or the ETC monitor trips fail-safe.

Spark Idle Load - Base [3×14] X=Coolant (°C) Y=Coolant (°C) range 0.135–0.5 [0xF74D8]

Spark Idle Load - Base Add | ECT Start LT -200 else 0 [1×7] X=(unused) range 0–0 [0xF53C8]

Data Integrity Checks - Spark Idle Load - PSP Comp

Description: ETC/torque safety-monitor plausibility (fail-safe). Keep in sync with the control maps.

Effect of changes: Raise together with the control map or the ETC monitor trips fail-safe.

PSP Comp Base Multiplier [1×8] X=Coolant (°C) range 0–1 [0xF5504]

Data Integrity Checks - Spark Idle Load - VSS Comp

Description: ETC/torque safety-monitor plausibility (fail-safe). Keep in sync with the control maps.

Effect of changes: Raise together with the control map or the ETC monitor trips fail-safe.

VSS Comp Multiplier | AT [3×8] X=Coolant (°C) Y=Coolant (°C) range 1–1.4 [0xF75AC]

VSS Comp Sub | In-Gear, MT [1×12] X=Engine RPM (rpm) range 0.0001–0.003 [0xF579C]

VSS Comp | Above VSS Threshold (5kph), AT [1×15] X=(unused) range 0–31.22 [0xF5808]

VSS Comp | Below VSS Threshold (5kph), AT [1×15] X=(unused) range 0–31.22 [0xF5880]

VSS Comp | In-Gear, MT [1×7] X=Vehicle Speed (km/h) range 0–0.1 [0xF5750]

Data Integrity Thresholds - DBW Throttle Duty Target

Description: Fail-safe decision thresholds. Normally leave alone.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Checksum Fault DTC P0601 - Throttle Duty Delta Threshold [29×12] X=Engine RPM (rpm) Y=Pedal (APP) (%) range 5.6–22 [0xF6084]

Checksum Fault DTC P0601 - Throttle Duty Delta Threshold | Cruise Control = 90 [0xF44AC]

Data Integrity Thresholds - DBW Torque Request

Description: Fail-safe decision thresholds. Normally leave alone.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Checksum Fault - Negative Torque Request Threshold [29×12] X=Engine RPM (rpm) Y=Pedal (APP) (%) range 2.48–199.2 [0xF6698]

Checksum Fault - Requested Torque Threshold [29×12] X=Engine RPM (rpm) Y=Pedal (APP) (%) range 1.36–160.12 [0xF5A70]

Checksum Fault - Requested Torque Threshold | Cruise Control = 2000 [0xF44A8]

Data Integrity Thresholds - DBW Torque Request

Description: Fail-safe decision thresholds. Normally leave alone.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

BYPASSED - WIP [29×12] X=Engine RPM (rpm) Y=Pedal (APP) (%) range 5.6–22 [0xF6CAC]

Emission System - Catalyst

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Catalytic Temp Estimate - Add | CL to OL Transition [1×9] X=Fuel range 0–220 [0xD4F68]

Catalytic Temp Estimate - Base [8×14] X=Engine RPM (rpm) Y=Engine Load range 359.4–882.6 [0xD4FE4]

Catalytic Temp Estimate - Multiplier | AFR Lean [8×14] X=Engine RPM (rpm) Y=Engine Load range 3.405–8.252 [0xD51FC]

Catalytic Temp Estimate - Multiplier | AFR Rich [8×14] X=Engine RPM (rpm) Y=Engine Load range 2.169–6.8 [0xD5414]

Emission System - EGR

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and

diagnostics).

EGR Target [13×10] X=Engine RPM (rpm) Y=Engine Load range 0–40 [0xC91E8]
EGR Target - 3rd Gear Sub (ZEROED) | AT [1×10] X=Engine RPM (rpm) range 0–0 [0xC906C]
EGR Target - 4th, 5th, 6th Gear Sub (ZEROED) | AT [1×10] X=Engine RPM (rpm) range 0–0 [0xC90A0]
EGR Target | IMRC [13×10] X=Engine RPM (rpm) Y=Engine Load range 0–40 [0xC9108]
EGR Target | Remote [13×10] X=Engine RPM (rpm) Y=Engine Load range 0–52 [0xC92C8]
WIP - EGR Target Calculation Multiplier | AFS Readiness Monitor Set [1×12] X=Intake Air (°C) range 40–100 [0xC93A4]

Emission System - EVAP

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

EVAPCP Base - Control Mod Voltage Add [1×9] X=Module V (V) range -1.5–24 [0xD5700]
EVAPCP Base - IAT Multiplier [1×15] X=Intake Air (°C) range 1–1.07 [0xD5760]
EVAPCP Base | Idle [10×15] X=(unused) Y=(unused) range 0–724 [0xD579C]
EVAPCP Base | Not Idle [10×15] X=(unused) Y=(unused) range 0–724 [0xD59F4]

Engine Fan Control

Description: Electric fan ON/OFF coolant thresholds.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

After Cooling - High Speed Fan Disable - Estimated ECT Threshold [1×8] X=Baro (kPa) range 45–45 [0xBF50]
After Cooling - High Speed Fan Enable - Estimated ECT Threshold [1×8] X=Baro (kPa) range 35–35 [0xBF20]
After Cooling - High Speed Fan Timer [1×8] X=Baro (kPa) range 0–0 [0xBFBB0]
After Cooling - Low Speed Fan Disable - Estimated ECT Threshold [1×8] X=Baro (kPa) range 186–186 [0xBFA90]
After Cooling - Low Speed Fan Enable - Estimated ECT Threshold [1×8] X=Baro (kPa) range 162–162 [0xBFA60]
After Cooling - Low Speed Fan Timer = 225 [0xBFA06]
After Cooling - Med Speed Fan Disable - Estimated ECT Threshold [1×8] X=Baro (kPa) range 96–96 [0xBFAF0]
After Cooling - Med Speed Fan Enable - Estimated ECT Threshold [1×8] X=Baro (kPa) range 84–84 [0xBFAC0]
After Cooling - Primary Med Speed Fan Timer [1×8] X=Baro (kPa) range 900–900 [0xBF80]
After Cooling - Secondary Med Speed Fan Timer = 750 [0xBFA08]
After Cooling Activation - Estimated ECT Threshold [1×8] X=Baro (kPa) range 115–115 [0xBFBE0]
Fan High Speed - ECT Activation Threshold | AC Off = 106 [0xBF9FC]
Fan High Speed - ECT Deactivation Threshold | AC Off = 113 [0xBF9FE]
Fan Low Speed to Disabled - VSS Threshold = 85 [0xBF1C]
Fan Low Speed to Disabled - VSS Threshold Hysteresis = 5 [0xBFA20]
Fan Mid Speed - ECT Activation Threshold | AC Off = 125 [0xBF9F8]
Fan Mid Speed - ECT Deactivation Threshold | AC Off = 132 [0xBF9FA]
Fan Mid Speed to Low Speed - VSS Threshold = 45 [0xBF14]
Fan Mid Speed to Low Speed - VSS Threshold Hysteresis = 5 [0xBF18]
Fan Selection - High Speed = 4368 [0xBFA04]
Fan Selection - Low Speed = 4096 [0xBFA00]
Fan Selection - Med Speed = 4112 [0xBFA02]
Fan Speed Transition - Delay = 120 [0xBF9F6]

Engine Limiters - RPM

Description: Rev/speed limiters.

Effect of changes: Raises/lowers rev or speed limiter. Mind mechanical limits.

Rev Limit Fuel Cut - Hysteresis A (ZERO) | 73C5 Delay Active = 0 [0xC4880]
Rev Limit Fuel Cut - Primary Low ECT Hysteresis = 5 [0xC47A8]
Rev Limit Fuel Cut - Primary Low ECT Threshold = -15 [0xC47A4]
Rev Limit Fuel Cut - Secondary Low ECT Hysteresis = 5 [0xC47B0]
Rev Limit Fuel Cut - Secondary Low ECT Threshold = -15 [0xC47AC]
Rev Limit Fuel Cut | AT w 73C0 Timer Active = 6500 [0xC4878]
Rev Limit Fuel Cut | AT w 73C0 Timer Expired, MT = 7500 [0xC487C]
Rev Limit Fuel Cut | ECT below Primary Threshold or Low/High Input Fault = 5500 [0xC4870]
Rev Limit Fuel Cut | ECT below Secondary Threshold, Start-Up = 5500 [0xC4874]
Rev Limit Throttle Cut | CPP clear = 10000 [0xC53C0]
Rev Limit Throttle Cut | CPP set = 6500 [0xC53BC]

Engine Limiters - VSS

Description: Rev/speed limiters.

Effect of changes: Raises/lowers rev or speed limiter. Mind mechanical limits.

Speed Limiter - RPM Activation Threshold = 4000 [0xC47E4]

Speed Limiter - RPM Activation Threshold Hysteresis = 93.75 [0xC47E8]

Speed Limiter - VSS Activation Threshold = 300 [0xC47EC]

Speed Limiter - VSS Activation Threshold Hysteresis = 5.64 [0xC47F0]

Engine Load - Limits

Description: Load metric scaling/limits (axis of most maps).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Load Limit | CMP Fault [1×13] X=Engine RPM (rpm) range 0.76–1.012 [0xC94F4]

Load Limit | No Fault [1×13] X=Engine RPM (rpm) range 0.76–1.032 [0xC955C]

Load Minimum - RPM Threshold = 900 [0xC941C]

Load Minimum - RPM Threshold Hysteresis = 100 [0xC9420]

Load Minimum | RPM Below Threshold [1×8] X=Coolant (°C) range 0.12–0.12 [0xC97B8]

Engine Load - Scaling

Description: Load metric scaling/limits (axis of most maps).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Load Scaling [17×14] X=Engine RPM (rpm) Y=Engine Load range 0.831528–1.2441 [0xCA1F0]

Load Scaling - VCT Comp Add | VCT GT 5 [15×14] X=Engine RPM (rpm) Y=Engine Load range 0–0.092 [0xCA61C]

Engine Sensors - AFS

Description: Sensor transfer functions. Adjust only when changing sensors/intake.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

AFS (Lambda) Scaling - Barometric Multiplier [1×4] X=Baro (kPa) range 1–1.4 [0xBF5C4]

AFS (Lambda) Scaling - Interpolated Current to Eq Ratio [1×11] X=EQ Ratio range 0.7586–1.3793 [0xBF600]

AFS (Lambda) Scaling - Non-Interpolated Current to Eq Ratio [1×11] X=EQ Ratio range 0.7586–1.3793 [0xBF658]

Engine Sensors - ECT

Description: Sensor transfer functions. Adjust only when changing sensors/intake.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

ECT Scaling - Voltage to ECT [1×39] X=Coolant (°C) range -40–150 [0xB8258]

Engine Sensors - IAT

Description: Sensor transfer functions. Adjust only when changing sensors/intake.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

IAT Scaling - Voltage to IAT [1×16] X=Intake Air (°C) range -40–119.38 [0xB8148]

Engine Sensors - KS

Description: Sensor transfer functions. Adjust only when changing sensors/intake.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

KS Scaling - Capture to Magnitude [1×55] X=Knock range 0–255 [0xBCB80]

KS Scaling - Minimum Proto-Knock Count for Successful Knock Event = 24 [0xBBCA7]

KS Scaling - Proto-Knock Magnitude Multiplier | Cylinder 1 [8×14] X=Engine RPM (rpm) Y=Engine Load range 2.35–3.4 [0xBDA44]

KS Scaling - Proto-Knock Magnitude Multiplier | Cylinder 2 [8×14] X=Engine RPM (rpm) Y=Engine Load range 2.35–3.4 [0xB

DCBC]

KS Scaling - Proto-Knock Magnitude Multiplier | Cylinder 3 [8×14] X=Engine RPM (rpm) Y=Engine Load range 2.35–3.4 [0x0BDED4]

KS Scaling - Proto-Knock Magnitude Multiplier | Cylinder 4 [8×14] X=Engine RPM (rpm) Y=Engine Load range 2.35–3.4 [0x0BE0EC]

Engine Sensors - KS - DO NOT MODIFY

Description: Sensor transfer functions. Adjust only when changing sensors/intake.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

KS - Capture End | Cylinder 1 [8×14] X=Engine RPM (rpm) Y=Engine Load range 448–576 [0x0BD194]

KS - Capture End | Cylinder 2 [8×14] X=Engine RPM (rpm) Y=Engine Load range 448–576 [0x0BD2CC]

KS - Capture End | Cylinder 3 [8×14] X=Engine RPM (rpm) Y=Engine Load range 448–576 [0x0BD404]

KS - Capture End | Cylinder 4 [8×14] X=Engine RPM (rpm) Y=Engine Load range 448–576 [0x0BD53C]

KS - Capture Start | Cylinder 1 [8×14] X=Engine RPM (rpm) Y=Engine Load range 51–192 [0x0BCCB4]

KS - Capture Start | Cylinder 2 [8×14] X=Engine RPM (rpm) Y=Engine Load range 51–192 [0x0BCDEC]

KS - Capture Start | Cylinder 3 [8×14] X=Engine RPM (rpm) Y=Engine Load range 51–192 [0x0BCF24]

KS - Capture Start | Cylinder 4 [8×14] X=Engine RPM (rpm) Y=Engine Load range 51–192 [0x0BD05C]

KS - DO NOT MODIFY [1×14] X=Engine RPM (rpm) range 0.4–0.4 [0x0BC97C]

KS - DO NOT MODIFY [1×15] X=(unused) range 0.1–1 [0x0BCA68]

KS - DO NOT MODIFY [8×14] X=Engine RPM (rpm) Y=Engine Load range 60–80 [0x0BD88C]

KS - DO NOT MODIFY [8×14] X=Engine RPM (rpm) Y=Engine Load range 12–18 [0x0BD674]

KS - DO NOT MODIFY - BYPASSED [1×14] X=Engine RPM (rpm) range 0.125–0.125 [0x0BC51C]

KS - DO NOT MODIFY - Max | Cylinder 1 [5×14] X=Engine RPM (rpm) Y=Engine Load range 28–33 [0x0BE2F8]

KS - DO NOT MODIFY - Max | Cylinder 2 [5×14] X=Engine RPM (rpm) Y=Engine Load range 28–33 [0x0BE45C]

KS - DO NOT MODIFY - Max | Cylinder 3 [5×14] X=Engine RPM (rpm) Y=Engine Load range 28–33 [0x0BE5C0]

KS - DO NOT MODIFY - Max | Cylinder 4 [5×14] X=Engine RPM (rpm) Y=Engine Load range 28–33 [0x0BE724]

KS - DO NOT MODIFY - Min | Cylinder 1 [5×14] X=Engine RPM (rpm) Y=Engine Load range 5–5 [0x0BE888]

KS - DO NOT MODIFY - Min | Cylinder 2 [5×14] X=Engine RPM (rpm) Y=Engine Load range 5–5 [0x0BE9EC]

KS - DO NOT MODIFY - Min | Cylinder 3 [5×14] X=Engine RPM (rpm) Y=Engine Load range 5–5 [0x0BEB50]

KS - DO NOT MODIFY - Min | Cylinder 4 [5×14] X=Engine RPM (rpm) Y=Engine Load range 5–5 [0x0BECB4]

KS - DO NOT MODIFY - Offset | Cylinder 1 [5×14] X=Engine RPM (rpm) Y=Engine Load range 20–25 [0x0BEE18]

KS - DO NOT MODIFY - Offset | Cylinder 2 [5×14] X=Engine RPM (rpm) Y=Engine Load range 20–25 [0x0BEF7C]

KS - DO NOT MODIFY - Offset | Cylinder 3 [5×14] X=Engine RPM (rpm) Y=Engine Load range 20–25 [0x0BF0E0]

KS - DO NOT MODIFY - Offset | Cylinder 4 [5×14] X=Engine RPM (rpm) Y=Engine Load range 20–25 [0x0BF244]

KS - DO NOT MODIFY - Proto-Knock Max Magnitude | Cylinder 1 [1×14] X=Engine RPM (rpm) range 319–319 [0x0BC58C]

KS - DO NOT MODIFY - Proto-Knock Max Magnitude | Cylinder 2 [1×14] X=Engine RPM (rpm) range 319–319 [0x0BC5FC]

KS - DO NOT MODIFY - Proto-Knock Max Magnitude | Cylinder 3 [1×14] X=Engine RPM (rpm) range 319–319 [0x0BC66C]

KS - DO NOT MODIFY - Proto-Knock Max Magnitude | Cylinder 4 [1×14] X=Engine RPM (rpm) range 319–319 [0x0BC6DC]

KS - DO NOT MODIFY - Proto-Knock Min Magnitude | Cylinder 1 [1×14] X=Engine RPM (rpm) range 64–64 [0x0BC74C]

KS - DO NOT MODIFY - Proto-Knock Min Magnitude | Cylinder 2 [1×14] X=Engine RPM (rpm) range 64–64 [0x0BC7BC]

KS - DO NOT MODIFY - Proto-Knock Min Magnitude | Cylinder 3 [1×14] X=Engine RPM (rpm) range 64–64 [0x0BC82C]

KS - DO NOT MODIFY - Proto-Knock Min Magnitude | Cylinder 4 [1×14] X=Engine RPM (rpm) range 64–64 [0x0BC89C]

Engine Sensors - MAF

Description: Sensor transfer functions. Adjust only when changing sensors/intake.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

DISABLED - MAF Scaling Alternate [1×129] X=MAF (g/s) range 0–213.6 [0x0C9E24]

High Load Corrected MAF Calc Exit - Load Threshold [1×13] X=Engine RPM (rpm) range 0.6–0.6 [0x0C9764]

High Load Corrected MAF Calc Exit - Throttle Duty Threshold [5×13] X=Engine RPM (rpm) Y=Cam Angle (°) range 22–40 [0x0CA070]

High Load Corrected MAF Calc Start - Transition Delay [1×13] X=Engine RPM (rpm) range 1–1 [0x0C9488]

High Load MAF Correction Factor = 0.861 [0x0C93E4]

MAF Scaling | Default [1×129] X=MAF (g/s) range 0–213.6 [0x0C9A1C]

MAF Scaling | Fault, Engine Off [1×8] X=Coolant (°C) range 0.6–1 [0x0C97F8]

Engine Sensors - MAP

Description: Sensor transfer functions. Adjust only when changing sensors/intake.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Estimated Differential Fuel Pressure | MAP Fault [15×12] X=Engine RPM (rpm) Y=Engine Load range 0.27–85.98 [0xB8854]
MAP Scaling - Added Offset (kpa) = 0 [0xB81A4]
MAP Scaling - Voltage to MAP (kpa) Multiplier = 24.71 [0xB81A8]

Fuel Comp CL - Decel Recovery

Description: Closed-loop fuel comps (short/long trims, cat checks).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Decel Recovery Enleanment Rate To Zero | Not OL, 7890 Expired = 3 [0xD3F80]
Decel Recovery Enrichment | Not OL, 789C Cleared [1×4] X=(unused) range 2–4 [0xD3FBC]
Decel Recovery Enrichment | Not OL, 789C Set [1×4] X=(unused) range 2.5–5 [0xD3F9C]

Fuel Comp CL - RO2S Emissions Monitor No Activity Check

Description: Closed-loop fuel comps (short/long trims, cat checks).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

RO2S Monitor No Activity Check - Enrich/Enlean Rate = 0.1 [0xD3F84]
RO2S Monitor No Activity Check - Enrichment Limit = 20.5 [0xD3F88]

Fuel Comp CL - RO2S Trim (ZERO)

Description: Closed-loop fuel comps (short/long trims, cat checks).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

RO2S Comp 77F0 - ZEROED Multiplier [1×14] X=MAF (g/s) range 0–0 [0xD40B4]
RO2S Comp 781C - ZEROED Enrichment | Delay 788A Not Expired = 0 [0xD3F48]

Fuel Comp CL - STFT Correction Coefficients

Description: Closed-loop fuel comps (short/long trims, cat checks).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Historic Eq Ratio Target Bias - Multiplier | Previous Lean [1×14] X=MAF (g/s) range 0.092–0.35 [0xD4700]
Historic Eq Ratio Target Bias - Multiplier | Previous Rich [1×14] X=MAF (g/s) range 0.1–0.44 [0xD4770]
STFT Correction MAF Coefficient A [1×14] X=MAF (g/s) range 0.2–1.4 [0xD45B0]
STFT Correction MAF Coefficient B [1×14] X=MAF (g/s) range 0.28932–0.891 [0xD4540]

Fuel Comp CL - STFT Transition Delay

Description: Closed-loop fuel comps (short/long trims, cat checks).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

STFT Transition Delay | Low ECT, Fault, Over Rich, Other WIP [1×4] X=Coolant (°C) range 30–77 [0xD4504]
STFT Transition Delay | Overfueling STFT, WIP = 110 [0xD4491]

Fuel Comp CL - Transition to Closed Loop

Description: Closed-loop fuel comps (short/long trims, cat checks).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Transition to CL - Delay [1×6] X=MAF (g/s) range 11–220 [0xD3D9C]
Transition to CL - Eq Ratio Delta (Act - Dsd) Threshold = 0.1 [0xD3D80]
Transition to CL - Fuel Enlean Limit = -25 [0xDAA60]
Transition to CL - Fuel Enrich Limit = 25 [0xDAA5C]
Transition to CL - Outside Eq Ratio Delta Threshold Delay = 4 [0xD3D71]

Fuel Comp CL - Warm Up Cat Efficiency Check RO2S Trim

Description: Closed-loop fuel comps (short/long trims, cat checks).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Warm-Up Cat Efficiency RO2S Trim - Base [1×22] X=O2 range -4.16–4.087 [0xD4024]

Warm-Up Cat Efficiency RO2S Trim - MAF Corrective Divisor [1×20] X=MAF (g/s) range 4.45–200.84 [0xD413C]

Warm-Up Cat Efficiency RO2S Trim - MAF Corrective Multiplier [1×20] X=MAF (g/s) range 0.200781–5.33223 [0xD41DC]

Fuel Comp CL - Warm-Up Cat Efficiency Check

Description: Closed-loop fuel comps (short/long trims, cat checks).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Warm-Up Cat Efficiency Check - CL Oscillation | P0421 Monitor Active, CL, Cruise, Cat @ Operating Temp [10×10] X=Engine RPM (rpm) Y=Engine Load range 5.1–12 [0xD49D8]

Warm-Up Cat Efficiency Check - Cat Temp Multiplier [1×8] X=Catalyst (°C) range 1–1 [0xD4668]

Warm-Up Cat Efficiency Check - ECT Multiplier [1×8] X=Coolant (°C) range 0.8–1 [0xD46A8]

Warm-Up Cat Efficiency Check - ZEROED | All Other Conditions [10×10] X=Engine RPM (rpm) Y=Engine Load range 0–0 [0xD47F8]

Fuel Comp OL - Corrective Spark Enrichment

Description: Open-loop fuel comps (decel cut, power enleanment, enrichment).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Corrective Spark Enrichment - Load Threshold = 0.85 [0xCCFEC]

Corrective Spark Enrichment - Load Threshold Hysteresis = 0.05 [0xCCFF0]

Corrective Spark Enrichment - RPM Threshold Hysteresis = 100 [0xCCFE8]

Corrective Spark Enrichment - RPM Threshold = 5000 [0xCCFE4]

Corrective Spark Enrichment Exit - Estimated Cat Temperature Threshold = 9999 [0xCD034]

Corrective Spark Enrichment | OL [1×7] X=Spark (°) range 0–0 [0xCD064]

Fuel Comp OL - Decel Fuel Cut

Description: Open-loop fuel comps (decel cut, power enleanment, enrichment).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Fuel Cut Multiplier - Enleanment Rate | Idle, In-Gear, TIme Active [6×5] X=Engine RPM (rpm) Y=Gear range 0.007–0.05 [0xCC774]

Fuel Cut Multiplier - Enleanment Rate | Not Idle Trans, In-Gear, Brake Off = 0.015 [0xCC698]

Fuel Cut Multiplier - Enleanment Rate | Not Idle Trans, In-Gear, Brake On = 0.007 [0xCC694]

Fuel Cut Multiplier - Enleanment Rate | Not Idle Trans, Not In-Gear, AT TIme Active = 0.001 [0xCC69C]

Fuel Cut Multiplier - Enleanment Rate | Not Idle Trans, Not In-Gear, MT or AT TIme Expired = 0.007 [0xCC6A0]

Fuel Cut Multiplier - Enleanment To Full Fuel Cut Threshold | 73F2 Mode 10 = 0.6 [0xCC690]

Fuel Comp OL - Decel Fuel Restore

Description: Open-loop fuel comps (decel cut, power enleanment, enrichment).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Decel Enrichment | Decel Mode 8, Brake Off, ECT GT 0, No Start-Up Catalyst Comps [1×13] X=Engine RPM (rpm) range 0–5 [0xCCE00]

Fuel Cut Multiplier - Enrichment From Starting Fuel Cut Value | 73F2 Mode 8, CPP Off = 0.688 [0xCC6A4]

Fuel Cut Multiplier - Enrichment Rate | 8250 Clear, 73D2 TIme Active, ECT Below 80 = 0.002 [0xCC6B4]

Fuel Cut Multiplier - Enrichment Rate | 8250 Clear, 73D2 TIme Expired, ECT Above 80 = 1 [0xCC6B8]

Fuel Cut Multiplier - Enrichment Rate | 8250 Set = 0.002 [0xCC6B0]

Fuel Cut Multiplier - Enrichment Rate | In-Gear, Brake Off = 0.003 [0xCC6AC]

Fuel Cut Multiplier - Enrichment Rate | In-Gear, Brake On = 0.003 [0xCC6A8]

Fuel Comp OL - Decel Fuel Restore Idle Thresholds

Description: Open-loop fuel comps (decel cut, power enleanment, enrichment).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Estimated Idle Speed Add A | Not In-Gear [1×8] X=Coolant (°C) range 1000–1500 [0xC4934]
Estimated Idle Speed Add B [1×8] X=Coolant (°C) range 218.75–1000 [0xC4974]
Estimated Idle Speed Add B | AT In-Gear, 73E9 On = 550 [0xC48D8]
Estimated Idle Speed Add B | In-Gear, AC On, Brake On, WIP = 170 [0xC48D0]
Estimated Idle Speed Add B | WIP = 850 [0xC48DC]
Estimated Idle Speed Add | RPM div VSS LT CCCC8, Gear GT 1, VSS GT CCCCC kph = 8 [0xC48F0]
Estimated Idle Speed Limiter - Stage 1 | PSP On, or PSP Off and AC On, 7F22 GT 0 = 1250 [0xC48E4]
Estimated Idle Speed Limiter - Stage 1 | PSP On, or PSP Off and AC On, 7F22 LTE 0 = 1531 [0xC48E0]
Estimated Idle Speed Limiter - Stage 2 | Brake On = 1600 [0xC48E8]
Limited Estimated Idle Speed Add | Prev 8250 Clear, 73E2 Clear and VSS LT 37.7 or 73E2 Set, AC Off, AT = 562.5 [0xC48AC]
Limited Estimated Idle Speed Add | Prev 8250 Clear, 73E2 Clear and VSS LT 37.7 or 73E2 Set, AC Off, MT = 562.5 [0xC48A8]
Limited Estimated Idle Speed Add | Prev 8250 Clear, 73E2 Clear and VSS LT 37.7 or 73E2 Set, AC On, AT = 1000 [0xC48B4]
Limited Estimated Idle Speed Add | Prev 8250 Clear, 73E2 Clear and VSS LT 37.7 or 73E2 Set, AC On, MT = 1000 [0xC48B0]
Limited Estimated Idle Speed Add | Prev 8250 Clear, 73E2 Clear, VSS GT 37.7, AC Off, AT = 312.5 [0xC48BC]
Limited Estimated Idle Speed Add | Prev 8250 Clear, 73E2 Clear, VSS GT 37.7, AC Off, MT = 562.5 [0xC48B8]
Limited Estimated Idle Speed Add | Prev 8250 Clear, 73E2 Clear, VSS GT 37.7, AC On, AT = 312.5 [0xC48C4]
Limited Estimated Idle Speed Add | Prev 8250 Clear, 73E2 Clear, VSS GT 37.7, AC On, MT = 1000 [0xC48C0]
Limited Estimated Idle Speed Add | Prev 8250 set, 8250 Clear, AC Off = 313 [0xC48A0]
Limited Estimated Idle Speed Add | Prev 8250 set, 8250 Clear, AC On = 1000 [0xC48A4]
RPM divided by VSS Threshold = 200 [0xC48EC]

Fuel Comp OL - Power Enleanment

Description: Open-loop fuel comps (decel cut, power enleanment, enrichment).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Power Enleanment - APP Threshold = 87.5 [0xCD01C]
Power Enleanment - APP Threshold Hysteresis = 5 [0xCD020]
Power Enleanment - Base | OL [13×14] X=Engine RPM (rpm) Y=Engine Load range 0–14 [0xCDBC4]
Power Enleanment - ECT Threshold = 80 [0xCCFF4]
Power Enleanment - ECT Threshold Hysteresis = 5 [0xCCFF8]
Power Enleanment - IAT Threshold = 15 [0xCD004]
Power Enleanment - IAT Threshold Hysteresis = 3 [0xCD008]
Power Enleanment - Load Threshold = 0.82 [0xCD038]
Power Enleanment - RPM Threshold = 6250 [0xCD024]
Power Enleanment - RPM Threshold Hysteresis = 500 [0xCD028]
Power Enleanment - Rate Limit = 1 [0xCD03C]
Power Enleanment - TP Threshold = 75 [0xCD014]
Power Enleanment - TP Threshold Hysteresis = 5 [0xCD018]
Power Enleanment Exit - ECT Threshold = 95 [0xCCFFC]
Power Enleanment Exit - ECT Threshold Hysteresis = 5 [0xCD000]
Power Enleanment Exit - IAT Threshold = 35 [0xCD00C]
Power Enleanment Exit - IAT Threshold Hysteresis = 3 [0xCD010]
Power Enleanment Exit - Rate Limit = 1 [0xCD040]
Power Enleanment Timer = 240 [0xCCFB8]
Power Enleanment Timer - Reset Delay = 40 [0xCCFB9]
Power Enleanment Timer - Reset Delay Activation Load Delta = 0.5 [0xCD044]
Power Enleanment Timer - Reset Delay Bypass Load Threshold [1×13] X=Engine RPM (rpm) range 0–1 [0xCD0B4]

Fuel Comp OL - Power Enleanment (OEM Bypassed)

Description: Open-loop fuel comps (decel cut, power enleanment, enrichment).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Power Enleanment (OEM Bypassed) - Enleanment Rate | OL = 0 [0xCD030]

Power Enleanment (OEM Bypassed) - Maximum | OL = 0 [0xCD02C]
 Power Enleanment (OEM Bypassed) - VSS Threshold = 999 [0xCCFBC]
 Power Enleanment (OEM Bypassed) - VSS Threshold Hysteresis = 5 [0xCCFC0]
 Power Enleanment (OEM Bypassed) Exit - ECT Threshold = 999 [0xCCFC4]
 Power Enleanment (OEM Bypassed) Exit - ECT Threshold Hysteresis = 5 [0xCCFC8]
 Power Enleanment (OEM Bypassed) Exit - Load Threshold = 999 [0xCCFCC]
 Power Enleanment (OEM Bypassed) Exit - Load Threshold Hysteresis = 0.05 [0xCCFD0]
 Power Enleanment (OEM Bypassed) Exit - RPM Threshold = 8000 [0xCCFD4]
 Power Enleanment (OEM Bypassed) Exit - RPM Threshold Hysteresis = 100 [0xCCFD8]

Fuel Comp OL - Start-Up Enrichment (Catalyst)

Description: Open-loop fuel comps (decel cut, power enleanment, enrichment).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Start-Up Enrichment A - Base | Engine OFF, 90 LT ECT LT 103, IAT GT 68 = 40 [0xCC974]
 Start-Up Enrichment A - Base | Engine OFF, 90 LT ECT LT 103, IAT LT 55 = 30 [0xCC96C]
 Start-Up Enrichment A - Base | Engine OFF, 90 LT ECT LT 103, IAT LT 68 = 35 [0xCC970]
 Start-Up Enrichment A - Base | Engine OFF, ECT GT 103, IAT GT 68 = 45 [0xCC980]
 Start-Up Enrichment A - Base | Engine OFF, ECT GT 103, IAT LT 55 = 35 [0xCC978]
 Start-Up Enrichment A - Base | Engine OFF, ECT GT 103, IAT LT 68 = 40 [0xCC97C]
 Start-Up Enrichment A - Base | Engine OFF, ECT LT 90 [1×8] X=Coolant (°C) range 12.5–120.313 [0xCC81C]
 Start-Up Enrichment A - Rate to Normal Fueling | Engine ON, 90 LT ECT LT 103, IAT LT 55 = 0.16 [0xCC9A4]
 Start-Up Enrichment A - Rate to Normal Fueling | Engine ON, ECT GT 103, IAT LT 55 = 0.16 [0xCC9B0]
 Start-Up Enrichment A - Rate to Normal Fueling | Engine ON, ECT LT 90 [1×8] X=Coolant (°C) range 0.2–0.44 [0xCC89C]
 Start-Up Enrichment B - Base | Engine OFF, 90 LT ECT LT 103, IAT GT 68 = 50 [0xCC98C]
 Start-Up Enrichment B - Base | Engine OFF, 90 LT ECT LT 103, IAT LT 55 = 30 [0xCC984]
 Start-Up Enrichment B - Base | Engine OFF, 90 LT ECT LT 103, IAT LT 68 = 45 [0xCC988]
 Start-Up Enrichment B - Base | Engine OFF, ECT GT 103, IAT GT 68 = 55 [0xCC998]
 Start-Up Enrichment B - Base | Engine OFF, ECT GT 103, IAT LT 55 = 50 [0xCC990]
 Start-Up Enrichment B - Base | Engine OFF, ECT GT 103, IAT LT 68 = 50 [0xCC994]
 Start-Up Enrichment B - Base | Engine OFF, ECT LT 90 [1×8] X=Coolant (°C) range 0–200 [0xCC85C]
 Start-Up Enrichment B - Rate to Normal Fueling | Engine ON, 90 LT ECT LT 103, IAT GT 68 = 1 [0xCC9BC]
 Start-Up Enrichment B - Rate to Normal Fueling | Engine ON, 90 LT ECT LT 103, IAT LT 55 = 1 [0xCC9B4]
 Start-Up Enrichment B - Rate to Normal Fueling | Engine ON, 90 LT ECT LT 103, IAT LT 68 = 1 [0xCC9B8]
 Start-Up Enrichment B - Rate to Normal Fueling | Engine ON, ECT GT 103, IAT GT 68 = 1 [0xCC9C8]
 Start-Up Enrichment B - Rate to Normal Fueling | Engine ON, ECT GT 103, IAT LT 55 = 1 [0xCC9C0]
 Start-Up Enrichment B - Rate to Normal Fueling | Engine ON, ECT GT 103, IAT LT 68 = 1 [0xCC9C4]
 Start-Up Enrichment B - Rate to Normal Fueling | Engine ON, ECT LT 90 [1×8] X=Coolant (°C) range 0.02441–1.3 [0xCC8DC]
 Start-Up Enrichment Multiplier | IMRC [1×8] X=Coolant (°C) range 1–1 [0xCC91C]
 Start-Up Enrichment Multiplier | IMRC [1×6] X=Idle range 0–0 [0xCC954]

Fuel Comp OL/CL - IMRC Transition

Description: Shared fuel comps (LTFT, transient enrichment).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

IMRC Transitional Enrichment Reset - TP Threshold [1×14] X=Engine RPM (rpm) range 999–999 [0xCE114]
 IMRC Transitional Enrichment Reset - TP Threshold Hysteresis [1×14] X=Engine RPM (rpm) range 0–0 [0xCE184]
 IMRC Transitional Enrichment Reset Value | 6EB8 Clear [1×8] X=Coolant (°C) range 0.4–0.8 [0xCE0BC]
 IMRC Transitional Enrichment Reset Value | 6EB8 Set [1×8] X=Coolant (°C) range 0–0 [0xCE07C]

Fuel Comp OL/CL - Increasing Load Enrichment

Description: Shared fuel comps (LTFT, transient enrichment).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

AC/PSP Activation Enrichment - Base [1×8] X=Coolant (°C) range 0–20 [0xCCF58]
 AC/PSP Activation Enrichment - Enleanment Rate To Zero [1×8] X=Coolant (°C) range 3.125–4 [0xCCF98]
 AC/PSP Activation Enrichment - Primary Activation Delay | PSP On = 0 [0xCCEF8]

AC/PSP Activation Enrichment - Secondary Activation Delay | PSP On [1×11] X=Coolant (°C) range 30–30 [0xCCF2C]
 Cold Start-Up - Increasing Load Enrichment Add Decrement Rate = 0.049 [0xCCD04]
 Cold Start-Up - Increasing Load Enrichment Add | Not Idle [1×8] X=Coolant (°C) range 0–35 [0xCCDAC]
 Increasing Load Enrichment - Transition From Idle Add | RPM Above Idle Speed, AT [1×8] X=Coolant (°C) range 0–11.88 [0xCCED8]
 Increasing Load Enrichment - Transition From Idle Decrement Rate | AT [1×8] X=Coolant (°C) range 0.39–5 [0xCCE98]
 Increasing Load Enrichment - Transition From Idle | AT [1×8] X=Coolant (°C) range 9.375–45 [0xCCE58]
 Increasing Load Enrichment A - Base | Load Delta Above Threshold, Not Idle [1×8] X=Coolant (°C) range 0–45 [0xCCD2C]
 Increasing Load Enrichment A - Load Delta Threshold | Timer 766C Active = 0.08 [0xCCCFF8]
 Increasing Load Enrichment A - Load Delta Threshold | Timer 766C Expired = 0.15 [0xCCCFF4]
 Increasing Load Enrichment B - Base | Load Delta and VSS Above Thresholds, Idle [1×8] X=Coolant (°C) range -20–0 [0xCCD6C]
 Increasing Load Enrichment B - Load Delta Threshold = 0.1 [0xCCD00]
 Increasing Load Enrichment B - VSS Threshold = 5 [0xCCD08]

Fuel Comp OL/CL - LTFT

Description: Shared fuel comps (LTFT, transient enrichment).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

LTFT - Aggregate Short Term Fuel Trim Accumulation Timer = 150 [0xD3DA3]
 LTFT - Maximum = 20 [0xD3E04]
 LTFT - Maximum EQ Ratio Act Delta to Set STFT Stable Flag | PSP Clear, 7714 Clear, 7719 Set = 0.015 [0xD3E10]
 LTFT - Maximum STFT Range to set LTFT Targeting Flag | STFT Stable Flag Set = 18 [0xD3E14]
 LTFT - Minimum = -20 [0xD3E08]

Fuel Comp OL/CL - LTFT MAF Breakpoints

Description: Shared fuel comps (LTFT, transient enrichment).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

LTFT MAF Breakpoint 0 - Idle = 5.7 [0xD3DB4]
 LTFT MAF Breakpoint 0 Hysteresis = 0.64 [0xD3DB8]
 LTFT MAF Breakpoint 1 = 15 [0xD3DBC]
 LTFT MAF Breakpoint 1 Hysteresis = 1 [0xD3DC0]
 LTFT MAF Breakpoint 2 = 30 [0xD3DC4]
 LTFT MAF Breakpoint 2 Hysteresis = 1.5 [0xD3DC8]
 LTFT MAF Breakpoint 3 = 200 [0xD3DCC]
 LTFT MAF Breakpoint 3 Hysteresis = 1.5 [0xD3DD0]
 LTFT MAF Breakpoint 4 = 200 [0xD3DD4]
 LTFT MAF Breakpoint 4 Hysteresis = 1.5 [0xD3DD8]
 LTFT MAF Breakpoint 5 = 200 [0xD3DDC]
 LTFT MAF Breakpoint 5 Hysteresis = 1.5 [0xD3DE0]

Fuel Comp OL/CL - MAF Thresholds WIP

Description: Shared fuel comps (LTFT, transient enrichment).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

RO2S Comp Deactivation - Max MAF Threshold = 22 [0xD3EB4]
 RO2S Comp Deactivation - Max MAF Threshold Hysteresis = 1 [0xD3EB8]

Fuel Comp OL/CL - Radiant Intake Heat

Description: Shared fuel comps (LTFT, transient enrichment).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Radiant Intake Heat Enrichment [7×13] X=Coolant (°C) Y=Engine Load range 0–10 [0xCDEF0]

Fuel IPW - Base

Description: Injector pulse width (fuel quantity). Set when changing injectors.

Effect of changes: Higher=richer, lower=leaner. Match injector characteristics when changed.

IPW Base = 10.492 [0xCB62C]

Fuel IPW - Cranking

Description: Injector pulse width (fuel quantity). Set when changing injectors.

Effect of changes: Higher=richer, lower=leaner. Match injector characteristics when changed.

IPW Cranking - Correction Multiplier | ECT Below 0, Idle [1×15] X=Engine RPM (rpm) range 0.125–1 [0xCACE4]

IPW Cranking - Fuel Cut APP Threshold = 87.5 [0xCAACC]

IPW Cranking - Fuel Cut APP Threshold Hysteresis = 2.734 [0xCAAD0]

IPW Cranking - Injector Pulse Width Target [1×15] X=Coolant (°C) range 6–85 [0xCAC2C]

Fuel IPW - Dynamic Trim Mult A (Large)

Description: Injector pulse width (fuel quantity). Set when changing injectors.

Effect of changes: Higher=richer, lower=leaner. Match injector characteristics when changed.

IPW Dynamic Trim Mult A | IMRC, RPM @ 1500 [5×8] X=Coolant (°C) Y=Engine Load range 0.4335–0.97 [0xCBAD8]

IPW Dynamic Trim Mult A | IMRC, RPM @ 2300 [5×8] X=Coolant (°C) Y=Engine Load range 0.621–0.98 [0xCBBAC]

IPW Dynamic Trim Mult A | IMRC, RPM @ 900 [5×8] X=Coolant (°C) Y=Engine Load range 0.36475–0.97 [0xCBA04]

IPW Dynamic Trim Mult A | RPM @ 1500 [5×8] X=Coolant (°C) Y=Engine Load range 0.4335–0.97 [0xCBD54]

IPW Dynamic Trim Mult A | RPM @ 2300 [5×8] X=Coolant (°C) Y=Engine Load range 0.621–0.98 [0xCBE28]

IPW Dynamic Trim Mult A | RPM @ 900 [5×8] X=Coolant (°C) Y=Engine Load range 0.36475–0.97 [0xCBC80]

Fuel IPW - Dynamic Trim Mult B (Small)

Description: Injector pulse width (fuel quantity). Set when changing injectors.

Effect of changes: Higher=richer, lower=leaner. Match injector characteristics when changed.

IPW Dynamic Trim Mult B | IMRC, RPM @ 1500 [5×8] X=Coolant (°C) Y=Engine Load range 0.04–0.19 [0xCBFD0]

IPW Dynamic Trim Mult B | IMRC, RPM @ 2300 [5×8] X=Coolant (°C) Y=Engine Load range 0.042–0.19 [0xCC0A4]

IPW Dynamic Trim Mult B | IMRC, RPM @ 900 [5×8] X=Coolant (°C) Y=Engine Load range 0.035–0.18 [0xCBEFC]

IPW Dynamic Trim Mult B | RPM @ 1500 [5×8] X=Coolant (°C) Y=Engine Load range 0.04–0.19 [0xCC24C]

IPW Dynamic Trim Mult B | RPM @ 2300 [5×8] X=Coolant (°C) Y=Engine Load range 0.042–0.19 [0xCC320]

IPW Dynamic Trim Mult B | RPM @ 900 [5×8] X=Coolant (°C) Y=Engine Load range 0.035–0.18 [0xCC178]

Fuel IPW - Dynamic Trim Mult C

Description: Injector pulse width (fuel quantity). Set when changing injectors.

Effect of changes: Higher=richer, lower=leaner. Match injector characteristics when changed.

IPW Dynamic Trim Mult C | AT with Timer Active [3×8] X=Coolant (°C) Y=Vehicle Speed (km/h) range 1–2 [0xCADC4]

IPW Dynamic Trim Mult C | MT or AT with Timer Expired [1×8] X=Coolant (°C) range 1.2–2.4 [0xCAC88]

Fuel IPW - Minimum

Description: Injector pulse width (fuel quantity). Set when changing injectors.

Effect of changes: Higher=richer, lower=leaner. Match injector characteristics when changed.

IPW Minimum [1×7] X=Coolant (°C) range 0.803–0.803 [0xCABA4]

Fuel IPW - Offset and Scaling

Description: Injector pulse width (fuel quantity). Set when changing injectors.

Effect of changes: Higher=richer, lower=leaner. Match injector characteristics when changed.

Injector Offset [9×16] X=Delta Y=Module V (V) range 0.243238–2.51524 [0xCB71C]

Injector Scaling - Cylinder 1 = 1 [0xCAAF8]

Injector Scaling - Cylinder 2 = 1 [0xCAAF8]

Injector Scaling - Cylinder 3 = 1 [0xCAB00]

Injector Scaling - Cylinder 4 = 1 [0xCAB04]

Injector Scaling - Differential Fuel Pressure [1×16] X=Delta range 0.890625–1.02148 [0xCB678]

Injector Scaling - Global = 1.065 [0xCAAD4]

Fuel Status OL (Primary) - APP

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and

diagnostics).

OL Transition (Primary) - APP Threshold = 82.031 [0xCAABC]
OL Transition (Primary) - APP Threshold Hysteresis = 2.734 [0xCAAC0]
OL Transition (Primary) - Delay | Low Baro = 60 [0xC4777]
OL Transition (Primary) - Delay | Normal Baro = 60 [0xC4776]

Fuel Status OL (Secondary) - APP

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

OL Transition (Secondary) - APP Threshold Hysteresis | Global = 2.734 [0xC489C]
OL Transition (Secondary) - APP Threshold | High Det [1×14] X=Engine RPM (rpm) range 75–75 [0xC4A3C]
OL Transition (Secondary) - APP Threshold | Normal [1×14] X=Engine RPM (rpm) range 75–75 [0xC4AAC]

Fuel Status OL (Secondary) - Load

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

OL Transition (Secondary) - Load Threshold - High ECT Activation Threshold = 200 [0xC4854]
OL Transition (Secondary) - Load Threshold - High ECT Activation Threshold Hysteresis = 5 [0xC4858]
OL Transition (Secondary) - Load Threshold Hysteresis | Global = 0.047 [0xC486C]
OL Transition (Secondary) - Load Threshold | High Det [1×14] X=Engine RPM (rpm) range 0–1 [0xC4BFC]
OL Transition (Secondary) - Load Threshold | High ECT [1×14] X=Engine RPM (rpm) range 0–1 [0xC4B8C]
OL Transition (Secondary) - Load Threshold | High ECT, High Det [1×14] X=Engine RPM (rpm) range 0–1 [0xC4B1C]
OL Transition (Secondary) - Load Threshold | Normal [1×14] X=Engine RPM (rpm) range 0–1 [0xC4C6C]

Fuel Status OL - Decel

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Decel Transition - Load Threshold (Highest Load to Hold Decel Status) | In-Gear [1×14] X=Engine Load range 0.01563–0.14063 [0xC49CC]
Decel Transition - Load Threshold Hysteresis = 0.016 [0xC4898]
Decel Transition - Load Threshold Timer Reset = 60 [0xC4775]
Decel Transition - RPM Threshold (Lowest RPM to Hold Decel Status) | In-Gear = 2312 [0xC47FC]
Decel Transition - RPM Threshold Count Required Enter Decel Status = 30 [0xC477D]
Decel Transition - RPM Threshold Hysteresis = 93.75 [0xC4800]

Fuel Status OL Delay Reset (Secondary)

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

OL Transition Delay Reset - BARO Below Threshold = 93 [0xC483C]
OL Transition Delay Reset - BARO Below Threshold Hysteresis = 0 [0xC4840]
OL Transition Delay Reset - ECT Threshold = 40 [0xC481C]
OL Transition Delay Reset - ECT Threshold Hysteresis = 5 [0xC4820]
OL Transition Delay Reset Value | BARO Below Threshold = 15 [0xC477B]
OL Transition Delay Reset Value | ECT Below Threshold = 15 [0xC4778]
OL Transition Delay Reset Value | Hi-Det = 15 [0xC4779]
OL Transition Delay Reset Value | Normal = 15 [0xC477A]

Fuel Status OL Delay Reset (Secondary) - Load

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

OL Transition Delay Reset (Secondary) - Max Load [1×14] X=Engine RPM (rpm) range 0.45313–0.45313 [0xC4D4C]
OL Transition Delay Reset (Secondary) - Max Load Hysteresis = 0.047 [0xC4910]

Fuel Status OL Delay Reset (Secondary) - RPM

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

OL Transition Delay Reset (Secondary) - Max RPM = 7969 [0xC4814]

OL Transition Delay Reset (Secondary) - Max RPM Hysteresis = 93.75 [0xC4818]

Fuel Status OL Delayed (Secondary) - Catalyst Temp

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

OL Transition Delayed (Secondary) - Max Estimated Cat Temp A = 850 [0xD4B8C]

OL Transition Delayed (Secondary) - Max Estimated Cat Temp B = 880 [0xD4B94]

OL Transition Delayed (Secondary) - Min Estimated Cat Temp A = 940 [0xD4B88]

OL Transition Delayed (Secondary) - Min Estimated Cat Temp B = 940 [0xD4B90]

Fuel Status OL Delayed (Secondary) - RPM

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

OL Transition Delayed (Secondary) - Max RPM Threshold = 5000 [0xC482C]

OL Transition Delayed (Secondary) - Max RPM Threshold Hysteresis = 93.75 [0xC4830]

OL Transition Delayed (Secondary) - Min RPM Threshold = 3289 [0xC4824]

OL Transition Delayed (Secondary) - Min RPM Threshold Hysteresis = 93.75 [0xC4828]

Fuel Status OL Delayed (Secondary) - TD

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

OL Transition Delayed (Secondary) - Max Throttle Duty Threshold [1×14] X=Engine RPM (rpm) range 0–60 [0xC4CDC]

OL Transition Delayed (Secondary) - Max Throttle Duty Threshold Hysteresis = 2.734 [0xC4908]

Fuel Target CL - Base (no patch required)

Description: Closed-loop target AFR (normally stoich/lambda 1).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Fuel Target CL - Base (Lambda) [13×10] X=Engine RPM (rpm) Y=Engine Load range 0.021–1.31732 [0xD4288]

Fuel Target CL - Base (requires patch)

Description: Closed-loop target AFR (normally stoich/lambda 1).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Fuel Target CL - Stoich AFR | Stock = 14.57, Gas = 14.7, E85 = 9.86 = 14.57 [0xD460C]

Fuel Target CL - Target AFR | Idle = 14.57 [0xD4610]

Fuel Target CL - Target AFR | Not Idle = 14.57 [0xD4614]

Fuel Target CL - Base Patch

Description: Closed-loop target AFR (normally stoich/lambda 1).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Fuel Target CL Patch | unpatched = 36620, patched = 9 = 36620 [0x4B58E]

Fuel Target CL Patch | unpatched = 36647, patched = 9 = 36647 [0x4B55C]

Fuel Target OL - Base

Description: Open-loop (high-load) target AFR. Rich=safe/power, lean=risky.

Effect of changes: Rich=safe/power, lean=economy but dangerous at high load.

Fuel Target OL (Lambda) | APP Above WOT Threshold, High Det [8×13] X=Engine RPM (rpm) Y=Throttle (TP) (%) range 13.1–36.1 [0xCD13C]

Fuel Target OL (Lambda) | APP Above WOT Threshold, Low Det [8×13] X=Engine RPM (rpm) Y=Throttle (TP) (%) range 13.1–36.1 [0xCD330]

Fuel Target OL (Lambda) | APP Below WOT Threshold, High Det [13×14] X=Engine RPM (rpm) Y=Engine Load range 0–36.1 [0xCD880]

Fuel Target OL (Lambda) | APP Below WOT Threshold, Low Det [13×14] X=Engine RPM (rpm) Y=Engine Load range 0–36.1 [0xCD53C]

Gear Ratio

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Gear Ratio - 1st = 3.815 [0xC135C]
Gear Ratio - 2nd = 2.26 [0xC1360]
Gear Ratio - 3rd = 1.64 [0xC1364]
Gear Ratio - 4th = 1.177 [0xC1368]
Gear Ratio - 5th = 1 [0xC136C]
Gear Ratio - 6th = 0.832 [0xC1370]
Gear Ratio - Final Drive = 4.1 [0xC1358]

Idle Speed - Base

Description: Target idle speed.

Effect of changes: Higher=faster idle, lower=risk of stall.

Idle Speed - Max = 2000 [0x3E724]

Idle Speed Comp

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Idle Speed Comp - Post Start-up Base - Decrement Rate = 2.5 [0xD6B08]
Idle Speed Comp - Post Start-up Base - Decrement Rate to Zero = 0.75 [0xD6B0C]
Idle Speed Comp - Post Start-up Base Limit - Decrement Rate (Ignored) = 2.5 [0xD6B10]
Idle Speed Comp - Post Start-up Base Limit - Decrement Rate To Zero (Ignored) = 0.75 [0xD6B14]

Idle Transition Delay

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Idle Transition Delay Reset - ECT Above Reset Threshold = 75 [0xCC6C8]
Idle Transition Delay Reset - ECT Above Reset Threshold Hysteresis = 5 [0xCC6CC]
Idle Transition Delay Reset - Gear Above Reset Threshold = 1.5 [0xCC6C4]
Idle Transition Delay Reset - VSS Above Reset Threshold = 20 [0xCC6D0]
Idle Transition Delay Reset - VSS Above Reset Threshold Hysteresis = 3.76 [0xCC6D4]
Idle Transition Delay Reset Value | Idle, Prev Not Idle, In-Gear, ECT, VSS, Gear Above Thresholds = 0 [0xCC682]
Idle Validation Switch - APP Threshold = 2.3 [0xBB674]
Idle Validation Switch - APP Threshold Hysteresis = 1 [0xBB678]
TD Transition Delay Reset - Throttle Duty Max Reset Threshold [1×14] X=Engine RPM (rpm) range 14–200 [0xCC710]
TD Transition Delay Reset Value = 0.031 [0xCC680]

Ignition Coil Dwell Time

Description: Coil charge time. Match to coil when changed.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Ignition Coil - Dwell Time | Speed Limit Inactive [9×13] X=Engine RPM (rpm) Y=Module V (V) range 2.15–9.5 [0xBBAD0]
Ignition Coil - Dwell Time | Speed Limiter Active = 0.768 [0xBB9FC]
Ignition Coil - Dwell WIP [1×14] X=Engine RPM (rpm) range 0–0.107 [0xBBA40]

Intake Manifold Runner Control (AT Only)

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

IMRC Cold Exit - ECT Threshold = 60 [0xC8D2C]
IMRC Cold Exit - ECT Threshold Hysteresis = 5 [0xC8D30]
IMRC Cold Exit - RPM Threshold = 3750 [0xC8D34]
IMRC Cold Exit - RPM Threshold Hysteresis = 93.75 [0xC8D38]
IMRC Cold Exit - Throttle Duty Threshold [1×14] X=Engine RPM (rpm) range 0–197.656 [0xC8D7C]
IMRC Cold Exit - Throttle Duty Threshold Hysteresis [1×14] X=Engine RPM (rpm) range 3.125–10 [0xC8DEC]
IMRC Transition - Decrement Rate Divisor [1×11] X=MAF (g/s) range 200–200 [0xBB8B0]

IMRC Transition - Increment Rate Divisor [1×11] X=MAF (g/s) range 1000–1000 [0xBB908]
 IMRC Transition - Increment and Decrement Rate = 2500 [0x59404]
 IMRC Transition - Interpolation Bias [1×5] X=2500/IMRC_DIVISOR_82DC range 0–1 [0xBB870]
 IMRC Warm Exit - Load Threshold [1×14] X=Engine RPM (rpm) range 0–0 [0xC8F3C]
 IMRC Warm Exit - Load Threshold Hysteresis [1×14] X=Engine RPM (rpm) range 0.1–0.1 [0xC8FAC]
 IMRC Warm Exit - RPM Threshold = 3250 [0xC8D3C]
 IMRC Warm Exit - RPM Threshold Hysteresis = 93.75 [0xC8D40]
 IMRC Warm Exit - Throttle Duty Threshold [1×14] X=Engine RPM (rpm) range 0–0 [0xC8E5C]
 IMRC Warm Exit - Throttle Duty Threshold Hysteresis [1×14] X=Engine RPM (rpm) range 3.125–3.125 [0xC8ECC]

Intake Manifold Tuning Valve

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

IMTV Open (Primary) - RPM Threshold = 4750 [0xBB978]
 IMTV Open (Primary) - RPM Threshold Hysteresis = 187.5 [0xBB97C]
 IMTV Open (Secondary) - Max RPM Threshold Hysteresis | Not Idle = 187.5 [0xBB974]
 IMTV Open (Secondary) - Max RPM Threshold | Not Idle = 3150 [0xBB970]
 IMTV Open (Secondary) - Throttle Duty Threshold [1×9] X=Engine RPM (rpm) range 15–110 [0xBB9A4]
 IMTV Open (Secondary) - Throttle Duty Threshold Hysteresis = 5 [0xBB964]

Patch - Decel Fuel Cutoff

Description: ROM code patches (immobilizer disable, displacement, DFCO, etc.). Handle with care.

Effect of changes: ROM modification; effect varies per item. Self-responsibility.

Patch - DFCO Disable | unpatched = 36610, patched = 9 = 36610 [0x45976]

Patch - Engine Displacement

Description: ROM code patches (immobilizer disable, displacement, DFCO, etc.). Handle with care.

Effect of changes: ROM modification; effect varies per item. Self-responsibility.

Patch - Engine Displacement - Patch Address A | patched = 1030656 = 824280 [0x3A728]
 Patch - Engine Displacement - Patch Address B | patched = 1030656 = 824280 [0x3AA90]
 Patch - Engine Displacement - Patch Address C | patched = 1030656 = 824280 [0x3AEE8]
 Patch - Engine Displacement | replace NaN value with engine displacement in cc = — [0xFBA00]

Patch - Immobilizer Disable

Description: ROM code patches (immobilizer disable, displacement, DFCO, etc.). Handle with care.

Effect of changes: ROM modification; effect varies per item. Self-responsibility.

Patch - Immobilizer Disable | unpatched = 36616, patched = 9 = 36616 [0x8F61E]

Patch - Open Loop Short Term Fuel Trim Fix

Description: ROM code patches (immobilizer disable, displacement, DFCO, etc.). Handle with care.

Effect of changes: ROM modification; effect varies per item. Self-responsibility.

Patch - Open Loop STFT Fix | unpatched = 141, patched = 160 = 141 [0x4C16E]

Patch - UDS Read Memory By Address (Mode 23)

Description: ROM code patches (immobilizer disable, displacement, DFCO, etc.). Handle with care.

Effect of changes: ROM modification; effect varies per item. Self-responsibility.

Patch - UDS Mode 23 Enable | unpatched = 0, patched = 16 = 0 [0x84C04]
 Patch - UDS Mode 23 Enable | unpatched = 36631, patched = 9 = 36631 [0x7F734]
 Patch - UDS Mode 23 Enable | unpatched = 6, patched = 15 = 6 [0x84C07]

Spark Base - High Fuel Request

Description: Base ignition timing (advance) vs RPM x load. Key map for power/economy/knock.

Effect of changes: Advance=more torque/economy but knock risk; retard=safest but less power. Most critical.

Spark Base - High Fuel Request | High-Det [15×14] X=Engine RPM (rpm) Y=Engine Load range -20–59.9 [0xCE7E4]
 Spark Base - High Fuel Request | High-Det, IMRC [15×8] X=Engine RPM (rpm) Y=Engine Load range -20–59.9 [0xCE590]

Spark Base - High Fuel Request | Low-Det [15×14] X=Engine RPM (rpm) Y=Engine Load range -12–59.9 [0xCF3D4]
Spark Base - High Fuel Request | Low-Det, IMRC [15×8] X=Engine RPM (rpm) Y=Engine Load range -12–59.9 [0xCF180]

Spark Base - High Fuel Request Transition

Description: Base ignition timing (advance) vs RPM x load. Key map for power/economy/knock.

Effect of changes: Advance=more torque/economy but knock risk; retard=safest but less power. Most critical.

Spark Base - High Fuel Request Threshold (Lambda) = 6.25 [0xCE390]
Spark Base - High Fuel Request Threshold Hysteresis (%) = 2.3 [0xCE394]
Spark Base - Low to High Fuel Request Transition Delay = — [0xCE38A]

Spark Base - Low Fuel Request

Description: Base ignition timing (advance) vs RPM x load. Key map for power/economy/knock.

Effect of changes: Advance=more torque/economy but knock risk; retard=safest but less power. Most critical.

Spark Base - Low Fuel Request | High-Det [15×14] X=Engine RPM (rpm) Y=Engine Load range -21–59.9 [0xCEDDC]
Spark Base - Low Fuel Request | High-Det, IMRC [15×8] X=Engine RPM (rpm) Y=Engine Load range -21–59.9 [0xCEB88]
Spark Base - Low Fuel Request | Low-Det [15×14] X=Engine RPM (rpm) Y=Engine Load range -12–59.9 [0xCF9CC]
Spark Base - Low Fuel Request | Low-Det, IMRC [15×8] X=Engine RPM (rpm) Y=Engine Load range -12–59.9 [0xCF778]

Spark Base / Idle Transition

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Transition to Idle Above Fuel Start-Up Enrichment Threshold = 6.6 [0xCE38C]
Transition to Idle Below RPM Threshold (additional requirement below) [1×6] X=Coolant (°C) range 4000–4000 [0xBF50]
Transition to Spark Base Above RPM Threshold [1×6] X=Coolant (°C) range 4500–4500 [0xBF20]

Spark Base Comp - ECT / IAT Combined

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

ECT / IAT Combined Comp - Advance Rate = 0.2 [0xCE398]
ECT / IAT Combined Comp - Base [9×9] X=Intake Air (°C) Y=Coolant (°C) range -6.25–14.5 [0xD00C4]
ECT / IAT Combined Comp - IMRC [9×9] X=Intake Air (°C) Y=Coolant (°C) range -6.25–14 [0xCFF38]
ECT / IAT Combined Comp - Power [9×9] X=Intake Air (°C) Y=Coolant (°C) range 0–0 [0xD0250]
ECT / IAT Combined Comp - RPM x Load Multiplier [7×14] X=Engine RPM (rpm) Y=Engine Load range 0.7–1.5 [0xCFD68]
ECT / IAT Combined Comp - Retard Rate to Zero = 0.2 [0xCE39C]

Spark Base Limit

Description: Advance-side spark limit (safety margin).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Spark Base Limit - VCT Add | VCT GT 5 [15×13] X=Engine RPM (rpm) Y=Engine Load range 0–4 [0xD0E44]
Spark Base Limit | High Fuel Request [15×14] X=Engine RPM (rpm) Y=Engine Load range -3.5–54.9 [0xD0850]
Spark Base Limit | IMRC [15×8] X=Engine RPM (rpm) Y=Engine Load range -3.5–54.9 [0xD0BF4]
Spark Base Limit | Low Fuel Request [15×14] X=Engine RPM (rpm) Y=Engine Load range -3.5–54.9 [0xD0494]

Spark Base Limit Comp - Power

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Power Comp - Advance Rate | Fuel Power Enleanment Set = 0 [0xD03A0]
Power Comp - Maximum Advance | Fuel Power Enleanment Set [15×14] X=Engine RPM (rpm) Y=Engine Load range 0–0 [0xD11C4]
Power Comp - Retard Rate to Zero | Fuel Power Enleanment Clear = 0 [0xD03A4]

Spark Comp - Cold Advance

Description: Ignition timing compensations (temperature, EGR, etc.).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

DISABLED - Spark Cold Advance | Not Idle [17×13] X=Coolant (°C) Y=Engine Load range 0-10 [0xD16C0]
Spark Cold Advance - Transition from Idle Add Retard Rate | Timer Expired = 0.7 [0xD1518]
Spark Cold Advance - Transition from Idle Add Timer = 0 [0xD150C]
Spark Cold Advance - Transition from Idle Add | Not Idle [1×7] X=Coolant (°C) range 0-5 [0xD162C]
Spark Cold Advance - ZEROED Add | Not Idle, IMRC, Above ECT Threshold [9×9] X=Engine RPM (rpm) Y=Engine Load range 0-0 [0xD2254]
Spark Cold Advance | Idle [1×7] X=Coolant (°C) range 0-10 [0xD1584]
Spark Cold Advance | Idle, IMRC [1×7] X=Coolant (°C) range 0-10 [0xD154C]
Spark Cold Advance | Not Idle [17×13] X=Coolant (°C) Y=Engine Load range -1-7 [0xD1AAC]
Spark Cold Advance | Not Idle, IMRC [17×13] X=Coolant (°C) Y=Engine Load range -1-7 [0xD1E98]

Spark Comp - EGR Advance

Description: Ignition timing compensations (temperature, EGR, etc.).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Spark EGR Advance [13×10] X=Engine RPM (rpm) Y=Engine Load range 0-15 [0xD2AAC]
Spark EGR Advance Actual to Target Ratio Multiplier [1×11] X=EGR Pos range 0-1 [0xD2770]
Spark EGR Advance | IMRC [13×10] X=Engine RPM (rpm) Y=Engine Load range 0-15 [0xD2848]
Spark EGR Advance | Remote Test Mode [13×10] X=Engine RPM (rpm) Y=Engine Load range 0-15 [0xD2F74]

Spark Correction - Feedback WIP (AT)

Description: Ignition corrections (knock, tip-in, cold, etc.).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Feedback Retard - Maximum | 6E53 Set, 6E5E Set = 39.727 [0xBFC68]

Spark Correction - High ECT Spark Corection WIP (AT Only)

Description: Ignition corrections (knock, tip-in, cold, etc.).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

WIP - High ECT Spark Corection - Index 0 | AT, Special = 1 [0xD2560]
WIP - High ECT Spark Corection - Index 1 | AT, Special = 10 [0xD2564]
WIP - High ECT Spark Corection - Index 2 | AT, Special = 20 [0xD2568]
WIP - High ECT Spark Corection - Index 3 | AT, Special = 10 [0xD256C]
WIP - High ECT Spark Corection - Index 4 | AT, Special = 5 [0xD2570]
WIP - High ECT Spark Corection - Index 5 | AT, Special = 1 [0xD2574]
WIP - High ECT Spark Corection - Index 6 | AT, Special = 0 [0xD2578]
WIP - High ECT Spark Corection - Index 7 | AT, Special = 0 [0xD257C]
WIP - High ECT Spark Corection - Index 8 | AT, Special = 0 [0xD2580]
WIP - High ECT Spark Corection - Index 9 | AT, Default (ZEROED) = 0 [0xD2584]

Spark Correction - High RPM and ECT

Description: Ignition corrections (knock, tip-in, cold, etc.).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Sarpk High RPM and ECT Retard Limit = 0 [0xD2660]
Spark High RPM and ECT Restore Advance Rate = 0 [0xD2664]
Spark High RPM and ECT Retard - Activation Delay = -0 [0xD2658]
Spark High RPM and ECT Retard - ECT Threshold = 100 [0xD2668]
Spark High RPM and ECT Retard - ECT Threshold Hysteresis = 5 [0xD266C]
Spark High RPM and ECT Retard - RPM Threshold = 5000 [0xD2670]
Spark High RPM and ECT Retard - RPM Threshold Hysteresis = 93.75 [0xD2674]
Spark High RPM and ECT Retard Rate = 0 [0xD265C]

Spark Correction - Knock Retard

Description: Ignition corrections (knock, tip-in, cold, etc.).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and

diagnostics).

High Det Mode - Minimum Accum KR Required = 11.25 [0xBBCE0]

High Det Mode - Minimum KR Required = 7777 [0xBB CDC]

KR Accumulator - Exit Delay = 0 [0xBB CA9]

KR Accumulator - Minimum KR for Accumulation [1×13] X=Engine RPM (rpm) range 2.8–5.625 [0xBB DF0]

Knock Detected - KR Increment Rate = 1.7 [0xBB CAC]

Knock Detected - KR Maximum [1×13] X=Engine RPM (rpm) range 8–8 [0xBB D20]

Knock Listen - Maximum RPM = 5250 [0xBB CC4]

Knock Listen - Maximum RPM Hysteresis = 93.75 [0xBB CC8]

Knock Listen - Minimum Load [1×13] X=Engine RPM (rpm) range 0.4–0.4 [0xBB D88]

Knock Listen - Minimum Load Hysteresis = 0.047 [0xBB CB8]

Knock Listen - Minimum RPM = 1000 [0xBB CBC]

Knock Listen - Minimum RPM Hysteresis = 93.75 [0xBB CC0]

Timing Restoration - KR Decrement Rate | Knock Listen Active [5×14] X=Engine RPM (rpm) Y=Engine Load range 0.17–0.17 [0xBB F98]

Timing Restoration - KR Decrement Rate | Knock Listen Inactive [5×14] X=Engine RPM (rpm) Y=Engine Load range 0.3515–0.3515 [0xBB C0FC]

Timing Restoration - KR Reduction Delay | Fast [5×14] X=Engine RPM (rpm) Y=Engine Load range 1–1 [0xBB F04]

Timing Restoration - KR Reduction Delay | Slow [5×14] X=Engine RPM (rpm) Y=Engine Load range 40–40 [0xBB E70]

Spark Correction - Stumble Recovery

Description: Ignition corrections (knock, tip-in, cold, etc.).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Spark Correction | MT, Not In-Gear, Idle, ECT Above 70, [1×10] X=Engine Load range 0–10 [0xCE2C8]

Spark Correction - Tip-In Anti Lug

Description: Ignition corrections (knock, tip-in, cold, etc.).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Tip-In Retard Anti Lug - Calculated Gear Alt WIP [1×12] X=Calc Gear (RPM/speed) range 2.55–6.29 [0xC045C]

Tip-In Retard Anti Lug - Calculated Gear | MT [1×12] X=Calc Gear (RPM/speed) range 1–6 [0xC048C]

Tip-In Retard Anti Lug - Current to Previous RPM Delta Multiplier [1×6] X=Gear range 0.5–1 [0xC0614]

Tip-In Retard Anti Lug - ECT Retard Multiplier [1×6] X=Coolant (°C) range 0–1 [0xC0534]

Tip-In Retard Anti Lug - Minimum RPM Threshold for Flag 7E73 Activation [1×6] X=Gear range 1400–1400 [0xC0644]

Tip-In Retard Anti Lug - Retard Limit [1×11] X=Engine RPM (rpm) range 0–20 [0xC0578]

Tip-In Retard Anti Lug - Retard Multiplier | Flag 7E66 Set, Flag 7E73 Clear [1×12] X=Counter range 0–1 [0xC04EC]

Tip-In Retard Anti Lug - Retard Multiplier | Flag 7E66 Set, Flag 7E73 Set [4×12] X=Counter Y=Pedal (APP) (%) range 0–1 [0xC069C]

Tip-In Retard Anti Lug Activation - Maximum Req Torque Threshold [1×11] X=Engine RPM (rpm) range -999–999 [0xC05D0]

Spark Correction - Tip-In First Gear

Description: Ignition corrections (knock, tip-in, cold, etc.).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Tip-In Retard First Gear - Add Advance | MT, AC On = 11.25 [0xBFF90]

Tip-In Retard First Gear - Base Retard | MT [1×10] X=Engine Load range 12–12 [0xBFFDC]

Tip-In Retard First Gear - IAT Multiplier | MT, AT [1×7] X=Intake Air (°C) range 0.75–0.99609 [0xC02A0]

Tip-In Retard First Gear - RPM Multiplier | MT [1×11] X=Engine RPM (rpm) range 0–0.96875 [0xC01A8]

Tip-In Retard First Gear - TP Multiplier | MT [1×9] X=Throttle (TP) (%) range 0–1 [0xC00C8]

Tip-In Retard First Gear - ZEROED Base Retard | AT, Flag 6E74 Clear [1×10] X=Engine Load range 0–0 [0xC007C]

Tip-In Retard First Gear - ZEROED Base Retard | AT, Flag 6E74 Set [1×10] X=Engine Load range 0–0 [0xC002C]

Tip-In Retard First Gear - ZEROED RPM Multiplier | AT, Flag 6E74 Clear [1×11] X=Engine RPM (rpm) range 0–0 [0xC0258]

Tip-In Retard First Gear - ZEROED RPM Multiplier | AT, Flag 6E74 Set [1×11] X=Engine RPM (rpm) range 0–0 [0xC0200]

Tip-In Retard First Gear - ZEROED Tip-In Multiplier | AT, Flag 6E74 Clear [1×9] X=Throttle (TP) (%) range 0–0 [0xC0158]

Tip-In Retard First Gear - ZEROED Tip-In Multiplier | AT, Flag 6E74 Set [1×9] X=Throttle (TP) (%) range 0–0 [0xC0110]

Spark Correction - Tip-In From Stop

Description: Ignition corrections (knock, tip-in, cold, etc.).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Tip-In Retard From Stop - Base Retard | AC Off = 11 [0xD7FF0]
Tip-In Retard From Stop - Base Retard | AC On = 6.5 [0xD7FEC]
Tip-In Retard From Stop - Multiplier Decrement to 0 | VSS GT 3.76 = 0.03 [0xD7FFC]
Tip-In Retard From Stop - Multiplier Decrement to 0 | VSS LT 3.76 = 0.02 [0xD7FF8]
Tip-In Retard From Stop - Multiplier Increment to 1 = 0.05 [0xD7FF4]

Spark Correction - Tip-In High Speed

Description: Ignition corrections (knock, tip-in, cold, etc.).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Tip-In Retard High Speed - APP Decreasing | AT [1×16] X=RPM (sub) (rpm) range 0–1.3 [0xC127C]
Tip-In Retard High Speed - APP Increasing | AT [1×16] X=RPM (sub) (rpm) range 0–1.3 [0xC11FC]
Tip-In Retard High Speed - ZEROED Base Retard | Gear 1st = 0 [0xC1108]
Tip-In Retard High Speed - ZEROED Base Retard | Gear 2nd = 0 [0xC1144]
Tip-In Retard High Speed - ZEROED Base Retard | Gear Other = 0 [0xC1180]
Tip-In Retard High Speed - ZEROED Base Retard | Unknown = 0 [0xC10CC]
Tip-In Retard High Speed - ZEROED Delay | Gear 1st [1×17] X=RPM (sub) (rpm) range 0–0 [0xC1008]
Tip-In Retard High Speed - ZEROED Delay | Gear 2nd [1×17] X=RPM (sub) (rpm) range 0–0 [0xC1060]
Tip-In Retard High Speed - ZEROED Delay | Gear Other [1×17] X=RPM (sub) (rpm) range 0–0 [0xC10B8]
Tip-In Retard High Speed - ZEROED Multiplier | VSS GT 125, ECT LT 100, ReqTq LT Threshold [1×16] X=RPM (sub) (rpm) range 0–1 [0xC12FC]

Spark Correction - Tip-In Shared

Description: Ignition corrections (knock, tip-in, cold, etc.).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Tip-In Retard Shared - ECT Multiplier | MT [1×13] X=Coolant (°C) range 0–0.9961 [0xC02F0]
Tip-In Retard Shared - ZEROED ECT Multiplier | AT, Flag 6E74 Clear [1×13] X=Coolant (°C) range 0–0 [0xC03C0]
Tip-In Retard Shared - ZEROED ECT Multiplier | AT, Flag 6E74 Set [1×13] X=Coolant (°C) range 0–0 [0xC0358]

Spark Correction - Transition from Decel

Description: Ignition corrections (knock, tip-in, cold, etc.).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Spark Transition from Decel - Base Retard | Idle, AC Off, AT [1×7] X=Coolant (°C) range 15–25 [0xD2520]
Spark Transition from Decel - Base Retard | Idle, AC Off, MT [1×7] X=Coolant (°C) range 15–15 [0xD24E8]
Spark Transition from Decel - Base Retard | Idle, AC On [1×7] X=Coolant (°C) range 15–20 [0xD24B0]
Spark Transition from Decel - Base Retard | Not Idle [1×9] X=Engine Load range 0–14.0625 [0xD23E0]
Spark Transition from Decel - Restore Advance Rate | 8250 Clear, AC On = 0.1 [0xD23A4]
Spark Transition from Decel - Restore Advance Rate | 7F22 Timer Active [1×9] X=Engine Load range 0.03–1.6 [0xD2428]
Spark Transition from Decel - Restore Advance Rate | 7F22 Timer Expired, MT [1×9] X=Engine Load range 0.2–1.6 [0xD2470]
Spark Transition from Decel - Restore Advance Rate | 8250 Clear, AC Off = 0.05 [0xD23A0]
Spark Transition from Decel - Restore Advance Rate | 8250 Set, AT = 0.05 [0xD23A8]

Spark Correction - Transition from In-Gear to Idle

Description: Ignition corrections (knock, tip-in, cold, etc.).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Spark Transition From In-Gear To Idle - Retard Limit | Idle [1×10] X=Engine Load range 0–20 [0xD25B8]
Spark Transition From In-Gear To Idle - Retard Multiplier | Prev In-Gear to Not In-Gear [1×7] X=Engine RPM (rpm) range 2–2.7 [0xD25FC]

Spark Transition From In-Gear To Idle - Retard Rate | Idle = 5.625 [0xD258C]
Spark Transition From In-Gear To Idle - Transition Retard Multiplier | Trans Timer Not Expired [1×8] X=Engine RPM (rpm) range 0–1 [0xD2638]
Spark Transition From In-Gear To Idle - Transition Timer Reset | Prev In-Gear to Not In-Gear = 0 [0xD2588]

Spark Idle / Base Limit - Transition

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Spark Idle - Transition Rate to Spark Base Limit = 0.03 [0xCE224]
Spark Idle - Transition Rate to Spark Idle | Fast = 0.02 [0xCE228]
Spark Idle - Transition Rate to Spark Idle | Slow [1×10] X=Engine RPM (rpm) range 0.01–1 [0xCE360]

Spark Idle / Base Limit Correction - Special Warm-Up WIP

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Correction Target Multiplier | In-Gear [1×17] X=Throttle (%) range 0–1 [0xD3918]
Correction Target Multiplier | Not In-Gear [1×17] X=Throttle (%) range 0–1 [0xD39A0]
Spark Idle Correction - Decrement Rate to Target | 7EF6 Clear, 7244 Set = 100 [0xD38AC]
Spark Idle Correction - Decrement Rate to Target | 7EF6 Clear, 7F0C Set = 0.2 [0xD38B4]
Spark Idle Correction - Decrement Rate to Target | 7EF6 Clear, Not Idle, In-Gear = 1.5 [0xD38B0]
Spark Idle Correction - Decrement Rate to Target | 7EF6 Set, Idle = 0.05 [0xD38BC]
Spark Idle Correction - Decrement Rate to Target | 7EF6 Set, Not Idle = 0.45 [0xD38B8]
Spark Idle Correction - Idle Speed Trim Multiplier [1×6] X=Idle range 0–1 [0xD3B1C]
Spark Idle Correction - Increment Rate To Target | Idle = 0.4 [0xD38C4]
Spark Idle Correction - Increment Rate To Target | Not Idle = 0.3 [0xD38C0]
Spark Idle Correction Target | In-Gear, 96F4 Set or Clear [1×9] X=Coolant (°C) range 0–0.1 [0xD3A50]
Spark Idle Correction Warm-Up Fuel Target | 7EF4 GT 0 [1×9] X=Coolant (°C) range 0–18 [0xD3AE0]
Spark Idle Correction ZEROED Target | In-Gear, 96F4 Set [1×9] X=Coolant (°C) range 0–0 [0xD3A08]
Spark Idle Correction ZEROED Target | Not In-Gear, 96F4 Set [1×9] X=Coolant (°C) range 0–0 [0xD3A98]

Spark Idle Limit

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Spark Idle Limit [9×11] X=Idle Y=Idle range -14.1–19 [0xD6428]
Spark Idle Limit | IMRC [9×11] X=Idle Y=Idle range -14.1–19 [0xD6604]

Spark Idle Limit Correction - AC Cycle Transition

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Spark Idle AC Cycle Off Comp - Duration | AC Cycle Off, Idle, Not In-Gear = 4 [0xD69C3]
Spark Idle AC Cycle Off Comp - Value | AC Cycle Off, Delay Active, In-Gear = 0 [0xD69F8]
Spark Idle AC Cycle Off Comp - Value | AC Cycle Off, Delay Active, Not In-Gear = 4 [0xD69FC]

Spark Idle Limit Correction - Idle Error

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Spark Idle Error Correction Cold - Base [1×11] X=Idle range -8.25–8.25 [0xD6864]
Spark Idle Error Correction Cold - ECT Base Alternative - Decrement Rate | 72CE Set, Not Decel, In-Gear = 0.3 [0xD6798]
Spark Idle Error Correction Cold - ECT Base Alternative - Decrement Rate | 72CE Set, Not Decel, Not In-Gear = 1 [0xD6794]
Spark Idle Error Correction Cold - ECT Base Alternative | In-Gear [1×8] X=Coolant (°C) range -10–0 [0xD6960]
Spark Idle Error Correction Cold - ECT Base Alternative | Not-In-Gear [1×8] X=Coolant (°C) range -20–-10 [0xD69A0]
Spark Idle Error Correction | In-Gear [1×11] X=Idle range -22.5–8 [0xD680C]
Spark Idle Error Correction | Not In-Gear, PS Off [1×11] X=Idle range -22.5–15 [0xD6914]
Spark Idle Error Correction | Not In-Gear, PS On [1×11] X=Idle range -22.5–15 [0xD68BC]

Spark Idle Limit Correction - PS Cycle Transition

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and

diagnostics).

Spark Idle PS Cycle Off Comp - Decrement Rate | PS Cycle On = 0.2 [0xD69F4]
Spark Idle PS Cycle Off Comp - Duration | PS Cycle Off, Idle = 14 [0xD69C1]
Spark Idle PS Cycle Off Comp - Value | PS Cycle Off, In-Gear, AC Off = 8 [0xD69EC]
Spark Idle PS Cycle Off Comp - Value | PS Cycle Off, In-Gear, AC On = 8 [0xD69E4]
Spark Idle PS Cycle Off Comp - Value | PS Cycle Off, Not In-Gear, AC Off = 5 [0xD69F0]
Spark Idle PS Cycle Off Comp - Value | PS Cycle Off, Not In-Gear, AC On = 8 [0xD69E8]
Spark Idle PS Cycle On Comp - Duration | PS Cycle On, Idle = 6 [0xD69C0]
Spark Idle PS Cycle On Comp - Value | PS Cycle On, In-Gear = 0 [0xD69CC]
Spark Idle PS Cycle On Comp - Value | PS Cycle On, Not In-Gear = 15 [0xD69D0]

Spark Idle Load - Alternator Compensation

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Alt Comp Decrement Rate | In-Gear, Pos Alt to Ctrl Mod Delta = 0.001 [0xD7814]
Alt Comp Decrement Rate | Not In-Gear, Pos Alt to Ctrl Mod Delta = 0.001 [0xD7818]

Spark Idle Load - PSP Compensation

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

PSP Comp Base Divisor Sub = 3 [0xD77B8]
PSP Comp Base Numerator and Divisor = 16 [0xD77BC]
PSP Comp Secondary Multiplier | PSP, AC, In-Gear = 0.03 [0xD7788]
PSP Comp Secondary Multiplier | PSP, AC, Not In-Gear = 0.05 [0xD778C]
PSP Comp Secondary Multiplier | PSP, In-Gear = 0.05 [0xD7798]
PSP Comp Secondary Multiplier | PSP, Not In-Gear = 0.08 [0xD779C]

Spark Limits

Description: Upper/lower spark-timing clamps (protection).

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Spark Limits - Default Ignition Timing | Fault = 9.86 [0xCE204]
Spark Limits - Max Advance = 50 [0xCE20C]
Spark Limits - Max Retard | Idle Stumble Recovery Correction Above 0 = -10 [0xCE220]
Spark Limits - Max Retard | RPM Above Idlespeed, Idle = -13.5 [0xCE218]
Spark Limits - Max Retard | RPM Below 700 or Above 8000, or ECT Below 20 -- Flag 7E66 Clear WIP = -10 [0xCE21C]
Spark Limits - Max Retard | RPM Below Idlespeed, Not Idle = -13.5 [0xCE210]
Spark Limits - Max Retard | RPM Below Idlespeed, Not Idle, Fault 71B0 set = -14.039 [0xCE214]
Spark Limits - Max Retard | Start-Up Timer Expired, Idle = -13.5 [0xCE230]
Spark Limits - Max Retard | Start-Up Timer Expired, Not Idle = -13.5 [0xCE234]
Spark Limits - Retard Limit | Flag 7ACC Clear (RPM below 3000 or not Idle) = -27.75 [0x4DD48]
Spark Limits - Retard Limit | Flag 7ACC Set (RPM below Idlespeed, WIP) [1×9] X=Coolant (°C) range -10--4 [0xCE314]

Spark To Cylinders

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Spark Final - Attack Rate Limiter [1×13] X=Engine RPM (rpm) range 5-5 [0xCE26C]
Spark Final - Cylinder 1 Trim = 0 [0xCE1F4]
Spark Final - Cylinder 2 Trim = 0 [0xCE1F8]
Spark Final - Cylinder 3 Trim = 0 [0xCE1FC]
Spark Final - Cylinder 4 Trim = 0 [0xCE200]

Variable Cam Timing - Base

Description: VVT target angle. Affects torque curve.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

VCT Activation - ECT Threshold = 63 [0xBA990]

VCT Activation - ECT Threshold Hysteresis = 5 [0xBA994]
VCT Target [15×13] X=Engine RPM (rpm) Y=Engine Load range 0–25 [0xBADC4]

Variable Cam Timing - DO NOT MODIFY

Description: VVT target angle. Affects torque curve.

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

CKP to CMP Synchronizing Offset [1×13] X=Engine RPM (rpm) range 9.581–11.603 [0xBAA28]
OCV Input for Commanded Duty [1×9] X=OCV range 1–4.6 [0xBAD30]
VCT Error - Correction to Apply | Above RPM Threshold [1×16] X=Cam Angle (°) range -600–400 [0xBAC8C]
VCT Error - Correction to Apply | Below RPM Threshold [1×16] X=Cam Angle (°) range -600–400 [0xBACCC]
VCT Error - ECT Multiplier [1×9] X=Coolant (°C) range 1–1 [0xBAC20]
VCT Error - Multiplier | Above RPM Threshold [1×13] X=Engine RPM (rpm) range 9–11 [0xBAA90]
VCT Error - Multiplier | Below RPM Threshold [1×13] X=Engine RPM (rpm) range 10–11 [0xBAAF8]
VCT Error - RPM Threshold = 7000 [0xBA998]
VCT Error - RPM Threshold Hysteresis = 93.75 [0xBA99C]
VCT Error Delta - ECT Multiplier [1×9] X=Coolant (°C) range 1–1 [0xBAC68]
VCT Error Delta - Multiplier | Above RPM Threshold [1×13] X=Engine RPM (rpm) range 0.0312–0.0625 [0xBAB60]
VCT Error Delta - Multiplier | Below RPM Threshold [1×13] X=Engine RPM (rpm) range 0.0312–0.0625 [0xBABC8]

WIP - Alternator

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Alternator - VSS Air-Flow Temperature Add [1×7] X=Vehicle Speed (km/h) range 7–16 [0xD84F4]
Alternator - Voltage Desired Add | [1×5] X=Alternator range -0.3–0.4 [0xD8684]
Alternator - Voltage Desired Add | RPM LT 5K, Load LT 0.80 [1×5] X=Alternator range -1.3–0 [0xD8634]
Alternator - Voltage Desired Base [1×9] X=Alternator range 12.85–14.72 [0xD8534]
Alternator Field Control Duty - Limiter | Fault [1×13] X=Engine RPM (rpm) range 0–100 [0xD88C0]
Alternator Field Control Duty | Fault [1×13] X=Alternator range 0–100 [0xD8858]
DISABLED WIP - Undefined [1×8] X=(unused) range 60–100 [0xD8758]
DISABLED WIP - Undefined [3×8] X=(unused) Y=(unused) range 40–100 [0xD9008]
Sensor_Scaling - Alt Amps to Sensor Voltage (not Batt V) [12×11] X=Alternator Y=Alternator range 0.3–4.98 [0xD9068]
Undefined [1×3] X=(unused) range 3–5 [0xD86EC]
Undefined [1×8] X=(unused) range 0–1 [0xD87D8]
Undefined [1×11] X=(unused) range 10–130 [0xD87F8]
Undefined [3×8] X=(unused) Y=(unused) range 90–100 [0xD8F7C]
WIP Base RPM Threshold [1×15] X=Coolant (°C) range 550–1100 [0xD83FC]
WIP Base RPM Threshold Hysteresis [1×15] X=Coolant (°C) range 250–300 [0xD8474]
WIP Limit | RPM below 312 or RPM LT Idle Speed [1×5] X=(unused) range 0.1–1 [0xD84C4]
WIP Timer [1×15] X=Coolant (°C) range 800–2800 [0xD83A0]
ZEROED [1×11] X=(unused) range 0–0 [0xD8920]

Warm-Up Fuel Target OL - Base

Effect of changes: Changing these values alters this category's behavior. Do not change if unsure (especially sensors and diagnostics).

Warm-Up Fuel Target - Decay Rate to Normal Fueling = 0.269 [0xD3C58]
Warm-Up Fuel Target - Dynamic Limiter (Lambda) [1×7] X=Coolant (°C) range 0–100 [0xCCC28]
Warm-Up Fuel Target - Dynamic Limiter (Lambda) | IMRC [1×7] X=Coolant (°C) range 0–100 [0xCCC60]
Warm-Up Fuel Target - Increasing Load Multiplier | In-Gear [1×13] X=Engine Load range 0.8–3.8 [0xCCA00]
Warm-Up Fuel Target - Increasing Load Multiplier | In-Gear, IMRC [1×13] X=Engine Load range 0.8–3.8 [0xCCA68]
Warm-Up Fuel Target - Increasing Load Multiplier | Not In-Gear [1×13] X=Engine Load range 0.95–4.75 [0xCCAD0]
Warm-Up Fuel Target - Increasing Load Multiplier | Not In-Gear, IMRC [1×13] X=Engine Load range 0.95–4.75 [0xCCB38]
Warm-Up Fuel Target - Increasing Load Secondary Multiplier | OL [1×13] X=Engine Load range 1–1.05263 [0xCCBA0]
Warm-Up Fuel Target - Multiplier A | IMRC [1×7] X=Coolant (°C) range 1–1 [0xCCC98]
Warm-Up Fuel Target - Multiplier B | IMRC [1×6] X=Idle range 0–0 [0xCCCCC]
Warm-Up Fuel Target OL (Lambda) [1×7] X=Coolant (°C) range 0–60 [0xCCBF0]

5. Map Interactions and Relationships

Maps are not independent: they interact through shared axes and through the ECU adding, subtracting and clamping their outputs. Changing one map changes how related maps behave, so understand the relationships before tuning.

5.1 Maps that share axes (changing one axis affects them all)

Several maps share the same axis (breakpoints). Changing that axis's scale moves the grid positions of every map that shares it at once. Main shared axes:

- Engine RPM (rpm)(RPM_6DD8): 145 maps share this — e.g. DTC Thresholds, Data Integrity Checks, Data Integrity Thresholds, Emission System, Engine Load
- Coolant (°C)(ECT_6D44): 72 maps share this — e.g. DBW, Engine Load, Engine Sensors, Fuel Comp CL, Fuel Comp OL
- Engine Load(LOAD_6F44): 66 maps share this — e.g. Emission System, Engine Sensors, Fuel Comp CL, Fuel Target CL, Spark Base
- Engine Load(LOAD_7004): 17 maps share this — e.g. Fuel Comp OL, Fuel Comp OL/CL, Fuel IPW, Fuel Status OL, Fuel Target OL
- Vehicle Speed (km/h)(VSS_DELTA_525C): 14 maps share this — e.g. DBW
- Pedal (APP) (%) (APP_6D1C): 13 maps share this — e.g. DBW, Data Integrity Checks, Data Integrity Thresholds
- Baro (kPa)(BARO_6DE8): 10 maps share this — e.g. Engine Fan Control, Engine Sensors
- Coolant (°C)(ECT_START_6D54): 10 maps share this — e.g. DTC Thresholds, Data Integrity Checks, Fuel Comp OL/CL
- Coolant (°C)(ECT_6D4C): 7 maps share this — e.g. Data Integrity Checks
- MAF (g/s)(MAF_W_IDLE/LOAD_LIMIT_6EF8): 7 maps share this — e.g. Fuel Comp CL

5.2 Ignition system interactions

Final ignition timing = 'Spark Base (base advance)' + various corrections, then clamped by the 'Spark Limits':

- 'Spark Base' = the base advance versus RPM x load (the starting point).
- 'Spark base corrections (ECT/IAT)' and 'spark corrections (knock retard, cold advance, tip-in, high-RPM / high-temp, etc.)' are added to or subtracted from the base.
- 'Spark Limits' cap the final value on the advance and retard sides.
- -> Even if you raise the base, the limit caps it and corrections shift the actual timing. Look at all three together.

5.3 Fuel system interactions

- Air mass ('engine sensor MAF/MAP' characteristics -> 'engine Load scaling') determines Load, which is the 'vertical axis' for ignition, fuel and VVT.
- Injection quantity = 'target AFR (Fuel Target OL/CL)' plus 'fuel injection pulse width (Fuel IPW base + offset / scaling)'.
- Then 'fuel corrections (LTFT/STFT, warm-up enrichment, acceleration enrichment, decel cut)' adjust it further.
- -> Changing the sensor characteristics or Load scaling shifts the lookup position of every map that uses Load as an axis (the widest-reaching change). When swapping injectors, always change the IPW base and offset together.

5.4 Intake and VVT interactions

- 'Variable intake (IMRC/IMTV)' switches at an RPM threshold and changes volumetric efficiency (VE) -> the fuel / ignition needed changes even at the same load.
- 'Variable valve timing (VVT)' also changes VE and internal EGR -> it affects the optimum fuel / ignition point.

5.5 Idle system interactions

'Target idle RPM' <-> 'Spark Idle' <-> 'idle load (air mass)' <-> 'idle corrections' work together to hold the idle. Tuning any one of them in isolation tends to make the idle unstable.

5.6 Limiters and integrity

- The 'rev limiter (fuel cut)' and 'throttle cut (clutch depressed)' are a condition-based pair; during a neutral free-rev the throttle cut acts first.
- Editing any map requires recalculating the 'checksum table (0xFF650+)' (corrected automatically on save).

- The 'flash counter (0xFFB00)' is a region the ECU updates on every write, so it is excluded from read-back comparison.